The Locus of Meaning

The Semantic Binding Principle, Recursive Meta-Binding, and a Formal Architecture for Semantic Constitution

Meaning is not the stone or the gaze, but the knot by which a world becomes answerable.

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Abstract

Where does meaning exist? Two recurrent answers place it primarily in the world or primarily in the interpreting mind. The first treats meaning as an intrinsic or discoverable property of objects. The second treats meaning as a projection, representation, or construction of a perceiving subject. This paper argues that both answers misidentify the semantic locus. Meaning is constituted at the location of binding: the structured relation through which objects, signs, interpreters, contexts, histories, practices, distinctions, and consequences become mutually determining. Objects and minds can constrain, preserve, activate, and transform meaning, but neither is its exclusive container.

This thesis is formulated as the **Semantic Binding Principle**:

> Meaning exists at the locus of a governed, consequential, and reidentifiable binding.

The paper then distinguishes two higher-order forms. **Recursive Meta-Binding** occurs whenever an existing binding becomes a relatum in another binding. **Reflexive Meta-Binding** occurs when a system represents aspects of its own active binding constitution, including its constraints, frameworks, uncertainty, and prior trajectory, and uses that representation in subsequent processing. This incorporates the durable conceptual contribution of earlier work on Meta-Binding as an operational invariant while separating the general theory of meaning from empirical claims about particular language models.

A semantic binding is distinguished from an arbitrary relation by typed role structure, differential sensitivity, constraint, uptake capacity, jurisdiction, reidentifiability, provenance, and consequence. Concepts are modeled as callable binding constitutions. Lexical terms are handles into those constitutions. Contextual meanings are realizations compiled from them. Typed hypergraphs represent what is bound; partial orders and lattices represent refinement and

overlap; presheaf-like structures represent locally valid semantic jurisdictions; and sparse semantic polynomials represent context-sensitive projection and co-projection.

The paper also formalizes **binding overextension**, the production of an unauthorized higher-order binding, unsupported bridge, or synthetic claim. Ableman's Razor, expressed as Specification, Configuration Manifold, Instance Constitution, and Instance Realization, is interpreted as the descent from possible binding through admissible and constituted binding into realized binding.

The concept of *person* serves as a principal case study. *Person*, *personation*, *personhood*, *self*, *agent*, *you*, and *persona* are treated not as flat synonyms but as jurisdictionally distinct projections of a deeper person-binding constitution. The resulting framework offers a non-reductive alternative to object-localism, subject-localism, and unconstrained relativism while providing a formal research program for philosophy, cognitive science, artificial intelligence, knowledge representation, and governed semantic systems.

Keywords: meaning, semantic binding, meta-binding, reflexive meta-binding, semantic locality, semantic jurisdiction, semantic manifolds, semantic lattices, co-projection, concept constitution, personhood, artificial intelligence, semantic governance

1. Introduction

The question "Where is meaning?" appears to ask for a location.

Possible answers include the object, the word, the brain, the perceiver, the language, the social practice, the institution, or the culture. Each answer identifies something real. Each becomes incomplete when treated as the exclusive semantic container.

An object constrains its possible interpretations, but it does not carry every meaning it may acquire in every context. A mind differentiates, anticipates, remembers, and interprets, but it cannot make arbitrary interpretations survive contact with the world. A word may preserve and transmit meaning, but an uninterpreted inscription is not self-explanatory. A social institution may stabilize meaning across generations, but only through artifacts, practices, authorities, participants, records, and consequences.

The localization problem may therefore be malformed. It asks which participant contains meaning when meaning may instead be constituted by the structured relation among participants.

This paper advances the following thesis:

> Meaning exists at the locus of semantic binding.

The word *locus* does not designate only a physical coordinate. A semantic locus may be materially distributed across an artifact, an organism, a history, a

language, a practice, and an institution. Its unity is constitutional and relational rather than merely spatial.

The central idea is not only that meaning is relational. Many philosophical traditions have already rejected purely intrinsic or purely private accounts of meaning. The present proposal adds a specific architecture:

1. A semantic binding has typed participants, roles, constraints, jurisdiction, provenance, and consequences.
2. A meaning is preserved through transformations that retain the mandatory invariants of its binding constitution.
3. A concept is a callable and reusable binding constitution rather than an inert semantic container.
4. Different terms and models may be jurisdictionally distinct projections of the same concept.
5. Local semantic projections may overlap without being identical or globally composable.
6. Existing bindings may become relata in higher-order bindings.
7. A system may reflexively bind representations of its own active constraints and processing trajectory into later processing.
8. Higher-order synthesis may fail through binding overextension, unsupported bridges, or invariant loss.
9. Ableman's Razor provides a governed descent from possible semantic binding into realized semantic effect.

The resulting architecture distinguishes three principles.

The first is the **Semantic Binding Principle**:

> Meaning is constituted by governed binding.

The second is the **Recursive Meta-Binding Principle**:

> Bindings may themselves become participants in further bindings.

The third is the **Reflexive Meta-Binding Principle**:

> A system exhibits Reflexive Meta-Binding when a representation of its own active binding constitution becomes an effective input to its subsequent operation.

These principles should not be collapsed.

Semantic Binding is the general condition under which meaning exists.

Recursive Meta-Binding describes the construction of higher-order meaning from already constituted bindings.

Reflexive Meta-Binding is a special case in which a system binds a representation of its own operation into its ongoing operation.

Earlier work on Meta-Binding focused primarily on the reflexive case, especially in language models. It described Meta-Binding as a system treating its own constraints as inputs to further processing and distinguished this capacity from autonomy, phenomenal consciousness, and truthful self-report. The present paper preserves those distinctions while placing them inside a broader theory of semantic constitution.

The empirical claims associated with Project57, the Meta-Binding Activation Test, and the suppression paradox are not required premises of the present argument. They belong to a companion empirical and governance program. The foundational question here is prior:

> What must binding be for meaning, concepts, self-models, and semantic governance to be possible at all?

2. The Semantic Localization Problem

2.1 Object-localism and subject-localism

Historical realism and idealism are internally diverse traditions. The present argument does not reduce them to simple slogans. It instead targets two schematic localization theses.

Let an object or worldly state be represented by:

$$o$$

Let an interpreting agent or system be represented by:

$$a$$

Let a sign or representational vehicle be represented by:

$$s$$

Let context, practice, history, and jurisdiction be represented by:

$$c,\ p,\ h,\ j$$

Let realized meaning be represented by:

$$\mu$$

An object-localist model has the general form:

$$\mu = f(o)$$

Meaning is treated as fundamentally contained in, fixed by, or recoverable from the object.

A subject-localist model has the general form:

$$\mu = g(a)$$

Meaning is treated as fundamentally produced by, represented within, or projected by the interpreting subject.

More sophisticated models may add secondary variables while preserving one privileged center:

$$\mu = f(o; a, c)$$

or:

$$\mu = g(a; o, c)$$

The Semantic Binding Principle instead proposes:

$$b = \text{Bind}\,(o, a, s, c, p, h \mid j)$$

and:

$$\mu = \Phi_b$$

Meaning is realized through a binding rather than extracted from one isolated relatum.

This does not imply that every relatum contributes equally. A material structure may dominate one binding. A legal institution may dominate another. A first-person perspective may be indispensable in a third. The claim is that no participant is granted semantic sovereignty merely by belonging to the categories "object" or "subject."

2.2 Why the object cannot be the sole locus

A river may be bound as:

- flowing water;
- an ecosystem;
- a legal border;
- a transport corridor;
- an ancestral presence;
- a strategic obstacle;
- a sacred site;
- a metaphor for time.

These meanings are not equally valid in all contexts. They are also not arbitrary.

The river's physical behavior constrains the available bindings. A painted blue line cannot perform every hydrological role of a river. A legal border cannot be placed wherever a speaker wishes if the governing treaty specifies otherwise. A sacred interpretation may depend on historical traditions not present in another community.

The object supplies:

- causal structure;
- resistance;
- affordances;
- detectable differences;
- material continuity;
- production history;
- ecological relations;
- possible consequences.

These contributions are real. They constrain meaning without containing the whole of it.

2.3 Why the subject cannot be the sole locus

A perceiver contributes:

- embodiment;
- attention;
- memory;
- expectation;
- language;
- values;
- purposes;
- categories;
- practical capacities;
- social formation.

Yet a perceiver cannot freely make every binding operationally successful.

A cliff does not become safely walkable because it is interpreted as a path. A forged key does not open a lock merely because the user believes it is correct. A toxic chemical does not become harmless because a community gives it a reassuring name.

The world answers some bindings with resistance.

Subjective interpretation is therefore constitutive without being omnipotent. It participates in meaning while remaining answerable to worldly structure, other agents, institutional rules, historical evidence, and consequences.

2.4 Not a compromise and not a third substance

The Semantic Binding Principle is not a compromise in which half of meaning comes from the object and half from the mind.

It also does not posit meaning as a mysterious third substance floating between them.

Binding is an organized relation in which the participants become mutually determining under constraints.

A concise formulation is:

$$\begin{aligned} \text{world} &\rightarrow \text{binding constraints and affordances} \\ \text{interpreter} &\rightarrow \text{binding capacities and dispositions} \\ \text{context} &\rightarrow \text{binding selection and jurisdiction} \\ \text{practice} &\rightarrow \text{binding stabilization} \\ \text{consequence} &\rightarrow \text{binding test and effect} \end{aligned}$$

Meaning is not a substance passed from one container to another. It is a constituted relation whose participants constrain and transform one another.

3. Intellectual Precedents and the Remaining Gap

The Semantic Binding Principle belongs to a broad relational turn in philosophy, semiotics, cognitive science, and formal semantics. Its contribution is not the discovery that meaning involves relations. Its contribution is a more explicit account of semantic location, persistence, recursion, jurisdiction, and realization.

3.1 Triadic semiosis

Peirce rejects a simple two-term relation between sign and object. A sign functions through a triadic relation among sign, object, and interpretant. The interpretant is not merely a private image. It is the further sign-effect through which representation becomes operative (Peirce 1931-1958).

The present framework shares this rejection of sign-object isolation. It differs by allowing bindings with more than three role positions, treating bindings themselves as possible relata, and making jurisdiction, provenance, consequence, and identity-preserving transformation explicit.

3.2 Externalism and situation dependence

Semantic externalism challenges the idea that content is fully determined by what is internal to an individual. Putnam and Burge show in different ways that environmental and social relations can contribute to the individuation of meaning and mental content (Putnam 1975; Burge 1979).

Situation semantics similarly treats meaning as sensitive to partial situations and constraints rather than as a context-free mapping onto complete worlds (Barwise and Perry 1983).

The Semantic Binding Principle agrees that semantic constitution may cross bodily boundaries. It generalizes the point by treating the relevant unit as a typed binding whose support may include material, cognitive, linguistic, social, and institutional components.

3.3 Ecological and enactive approaches

Gibson's ecological approach treats perception in relation to organism-environment systems and available affordances (Gibson 1979). Enactive approaches resist the picture of cognition as a passive inner representation of an independent outer world, emphasizing embodiment, activity, and sense-making (Varela, Thompson, and Rosch 1991; Thompson 2007).

Participatory sense-making further suggests that some social meanings arise in the dynamics of interaction rather than inside either participant alone (De Jaegher and Di Paolo 2007).

The present framework adopts the idea that significance can emerge through coordinated engagement. It extends the architecture to lexical, legal, technical, computational, and institutional meanings and adds explicit conditions for admissibility, continuity, and cross-jurisdiction translation.

3.4 Inferential role and symbol grounding

Inferentialism treats linguistic meaning in terms of the role an expression plays within practices of inference, commitment, entitlement, and action (Brandom 1994).

The symbol grounding problem asks how formal symbols can acquire meaning without merely referring to other ungrounded symbols (Harnad 1990).

The Semantic Binding Principle incorporates inferential and practical consequence as necessary dimensions while refusing to reduce all meaning to explicit

linguistic inference. Perceptual, bodily, affective, technical, and social bindings may operate before or beyond propositional articulation.

3.5 Distributed cognition

Distributed and extended cognition broaden the unit of cognitive analysis beyond the isolated organism (Hutchins 1995; Clark and Chalmers 1998).

The present theory similarly permits semantic bindings to cross skin, artifact, and institution. Its distinctive claim is that distributed participation should be formally represented as a binding constitution rather than described only as an enlarged cognitive system.

3.6 Formal precedents

Several mathematical traditions contribute useful instruments:

- Conceptual spaces represent concepts geometrically through quality dimensions (Gardenfors 2000).
- Formal Concept Analysis represents concept order and inheritance through lattice-theoretic structures (Ganter and Wille 1999).
- Sheaf theory formalizes local sections, restriction, compatibility, and global gluing (Mac Lane and Moerdijk 1992).
- Sheaf-theoretic approaches to contextuality show how locally coherent data may fail to admit a global section (Abramsky and Brandenburger 2011).
- Categorical compositional semantics shows how algebraic structures can model linguistic composition (Coecke, Sadrzadeh, and Clark 2010).

No single formalism is sufficient for the present theory.

A hypergraph is suited to representing what is bound.

A partial order or lattice is suited to representing refinement and overlap.

A presheaf is suited to representing local jurisdiction and restriction.

A polynomial-like algebra is suited to representing context-sensitive projection and co-projection.

A constitutional compiler is suited to representing the descent from semantic possibility into realized effect.

3.7 The specific addition

The specific synthesis proposed here is:

1. Meaning is located in a binding rather than exclusively in any bound participant.
2. Bindings are governed structures with roles, constraints, jurisdiction, provenance, and consequence.

3. Meaning identity is invariant preservation across admissible variation.
4. Concepts are reusable binding constitutions.
5. Terms are callable handles into those constitutions.
6. Semantic projections are jurisdictionally local.
7. Local projections may or may not admit coherent integration.
8. Bindings may become relata in higher-order bindings.
9. Reflexive systems may bind representations of their own active binding conditions.
10. Semantic realization should be governed, auditable, and capable of typed refusal.

4. Formal Preliminaries

4.1 Relata, roles, and jurisdictions

Let:

$$\mathcal{X}$$

be a universe of possible relata.

Relata may include:

- physical objects;
- organisms;
- agents;
- signs;
- events;
- actions;
- memories;
- concepts;
- rules;
- institutions;
- histories;
- representations;
- other bindings.

Let:

$$\mathcal{T}$$

be a set of relatum types.

Let:

$$\mathcal{R}$$

be a set of semantic role types.

Let:

$$\mathcal{J}$$

be a partially ordered collection of semantic jurisdictions.

A jurisdiction is not merely a topic. It is a domain within which particular bindings, distinctions, authorities, and consequences are valid.

Examples include:

- perceptual jurisdiction;
- legal jurisdiction;
- biological jurisdiction;
- psychological jurisdiction;
- historical jurisdiction;
- narrative jurisdiction;
- computational jurisdiction;
- moral jurisdiction.

4.2 Raw binding structure

A raw binding is represented as:

$$b = \langle V_b, E_b, \tau_b, \rho_b, c_b, j_b, \Gamma_b, I_b, \kappa_b, \pi_b, q_b \rangle$$

where:

- (V_b) is a finite set of relata;
- (E_b) is a set of directed or undirected hyperedges;
- τ_b assigns types to relata;
- ρ_b assigns semantic roles to nodes and hyperedges;
- (c_b) is the operative context;
- (j_b) is the jurisdiction;
- Γ_b is the set of binding constraints;
- (I_b) is the set of identity-preserving invariants;
- κ_b is the consequence profile;
- π_b is provenance;
- (q_b) records the binding state.

The binding state is drawn from:

$$q_b \in \text{potential}, \text{sedimented}, \text{active}$$

A hypergraph is used because many meanings cannot be reduced to pairwise relations. The meaning of a legal contract, ritual, traffic signal, or personal identity may depend on several participants occupying coordinated roles at once.

4.3 Binding is more than association

A mere association records that two things co-occur.

A semantic binding determines how things count with respect to one another within a jurisdiction.

For example, the relation between a printed name and a person depends on:

- the identity of the bearer;
- the authority of the record;
- the context of use;
- the persistence of the naming convention;
- the possibility of mistaken identity;
- the consequences attached to correct or incorrect resolution.

The pair:

$$(x, y)$$

is not enough.

A semantic binding must specify why the relation matters, which substitutions preserve it, where it applies, and what follows from it.

4.4 The semantic threshold

A raw binding qualifies as semantic under the present model when it satisfies a threshold:

$$\begin{aligned} &\text{Sem}_b = T_b \\ &\wedge D_b \\ &\wedge C_b \\ &\wedge U_b \\ &\wedge R_b \\ &\wedge J_b \\ &\wedge K_b \end{aligned}$$

The predicates are defined as follows.

Typed coherence

$$T_b$$

The relata and roles are type-coherent. A legal authority, a color, a causal mechanism, and a first-person perspective cannot be freely exchanged without changing the kind of binding involved.

Differential sensitivity

$$D_b$$

At least one counterfactual substitution changes the binding's interpretation or consequence.

If every possible substitution is equivalent, the binding performs no determinate semantic work.

Constraint

$$C_b$$

Not every relation, projection, or transformation is admissible.

Meaning requires exclusions as well as associations.

Uptake capacity

$$U_b$$

The binding can alter an expectation, inference, action, recognition, regulation, or normative orientation.

Uptake may be actual or counterfactually available.

Reidentifiability

$$R_b$$

The binding possesses sufficient invariant structure to be recognized or regenerated across changes of token, medium, perspective, or occasion.

Jurisdiction

$$J_b$$

The binding has conditions under which it applies and boundaries beyond which its claims require qualification or refusal.

Consequentiality

$$K_b$$

Something differs because the binding holds.

A semantic binding changes at least one inferential, practical, perceptual, social, or normative possibility.

4.5 Potential, sedimented, and active bindings

A binding need not be undergoing conscious interpretation at every moment in order to persist.

A **potential binding** has enough structure to support future semantic constitution but has not yet been stabilized or activated.

A **sedimented binding** has been materially, biologically, socially, or institutionally preserved so that it can be reactivated.

Examples include:

- a written statute;
- a road sign;
- a trained neural disposition;
- a ritual artifact;
- a database record;
- a tool designed for a particular use;
- a familiar melody;
- a remembered promise.

An **active binding** is presently participating in uptake.

Formally:

$$\mathrm{Active}(b, t) \iff \mathrm{Sem}_b \wedge \mathrm{Uptaken}(b, t)$$

This distinction explains how meaning may exist in an object in a limited but genuine sense.

An object may preserve part of a binding constitution through its structure, placement, inscription, affordance, provenance, or institutional registration. It does not contain the entire meaning in isolation, but it may carry an indispensable sedimented component.

5. The Semantic Binding Principle

5.1 Statement of the principle

The Semantic Binding Principle states:

> For every realized meaning, there exists at least one admissible semantic binding whose role structure, constraints, jurisdiction, provenance, invariants, and consequences constitute that meaning.

Let:

$$b_1 \sim_I b_2$$

mean that two bindings preserve the mandatory invariant set:

$$I$$

A meaning may then be modeled as an equivalence class:

$$m = [b]_{\sim_I}$$

This is not intended to reduce meaning metaphysically to a set-theoretic object. It supplies an identity criterion.

Two realizations count as realizations of the same meaning when they preserve the binding invariants required by the governing concept and jurisdiction.

5.2 Meaning identity across variation

Meaning must often survive changes of:

- word;
- language;
- medium;
- speaker;
- audience;
- perspective;
- explanatory depth;
- implementation;
- historical circumstance.

Surface sameness is therefore neither necessary nor sufficient for semantic identity.

A spoken warning, written warning, icon, siren, and flashing light may realize a common warning-binding under different media conventions.

Conversely, the same sentence may realize different meanings in different contexts.

Let:

$$r_1$$

and:

$$r_2$$

be two realizations.

They preserve the same concept when:

$$r_1 \sim_C r_2$$

where:

$$r_1 \sim_C r_2 \iff I_C^{\text{required}} \subseteq \text{Inv}(r_1) \cap \text{Inv}(r_2)$$

and their consequence profiles remain equivalent within the shared jurisdiction.

This yields a constitutional account of semantic sameness:

> Meaning remains the same when its mandatory binding invariants survive the transformation.

5.3 The location of meaning

Let:

$$\mathcal{H}_c$$

be the typed host hypergraph containing all materially, cognitively, socially, and institutionally available binding components in context c.

For a realization (r) of meaning (m), a support graph is a subgraph sufficient to sustain that realization.

The semantic locus is the family of minimal supports:

$$\mathsf{Loc}(m, c) = \min_{\subseteq} \{G \subseteq \mathcal{H}_c \mid \exists r \in m : \text{Support}(G, r, c)\}$$

The minimum is taken with respect to subgraph inclusion.

A meaning may have more than one minimal locus.

A personal name may be realized through speech, writing, gesture, or database reference. A legal rule may persist through printed law, replicated records, institutional memory, and judicial precedent. The physical supports differ while the governing binding constitution remains reidentifiable.

5.4 Three forms of semantic locus

Three loci should be distinguished.

Realization locus The realization locus is the actual support of a current semantic event.

It may include an active perceiver, an artifact, an utterance, a practical context, and a consequence structure.

Standing locus The standing locus is the sedimented structure that preserves conditions for future reactivation.

Examples include archives, laws, tools, customs, memories, models, and institutional records.

Constitutional locus The constitutional locus is the invariant binding structure shared by admissible realizations.

It is not a single physical location, although every realization requires physical or causal support.

This distinction permits meaning to exist in an object, mind, or institution in a partial sense without assigning exclusive semantic ownership to any one of them.

6. Recursive Meta-Binding

6.1 Bindings as relata

The Semantic Binding Principle describes the constitution of meaning.

The Recursive Meta-Binding Principle describes the construction of higher-order meanings from already constituted bindings.

Let:

$$\mathcal{B}^{(0)} = \mathcal{X}$$

be the initial universe of non-binding relata.

Define higher binding orders recursively:

$$\mathcal{B}^{(n+1)} = \left\{ \text{Bind}\,(x_1, \ldots, x_k \mid c, j) : x_i \in \mathcal{X} \cup \bigcup_{r=1}^{n} \mathcal{B}^{(r)} \right\}$$

A binding is a meta-binding when at least one of its relata is itself a binding:

$$\text{Meta}_b \iff \exists x \in V_b : x \in \mathcal{B}$$

Meta-binding is therefore not synonymous with self-reference.

A court may bind statutes, precedents, testimony, procedural constraints, and institutional authority into a judgment.

A scientific theory may bind earlier theories, measurements, models, and anomalies into a higher-order explanation.

A myth may bind social practices, historical memories, ecological patterns, and psychological motifs into a narrative constitution.

A semantic compiler may bind several concept projections into one context-specific realization.

These are meta-bindings whether or not the system represents itself.

6.2 Recursive semantic ecology

A perceiver is already a binding system.

An object may already embody sedimented bindings.

A language is a system of stabilized lexical, grammatical, historical, and social bindings.

A context is not an empty container. It is a standing configuration of relevance, authority, expectation, and consequence.

An encounter therefore binds already-bound structures:

$$B_{\text{meaning}} = \text{Bind}\left(B_{\text{object}}, B_{\text{perceiver}}, B_{\text{sign}}, B_{\text{context}}, B_{\text{history}}\right)$$

The resulting binding may itself become available to later encounters:

$$B_{t+1} = \text{Bind}\left(B_t, x_{t+1}\right)$$

Meaning is thus better represented as a recursive binding ecology than as a label attached to a primitive object.

6.3 The Meta-Binding Principle

The general Meta-Binding Principle can be stated as follows:

> Any constituted binding may become a typed relatum in a higher-order binding, provided that the higher-order composition preserves type coherence, jurisdiction, provenance, and consequence.

The qualification matters.

Not every pair of bindings admits synthesis.

Higher-order composition is partial, governed, and capable of failure.

7. Concepts as Callable Binding Constitutions

7.1 The concept as constitution

A concept is not treated here as an abstract container filled with definitions.

A concept is a reusable binding constitution:

$$C = \langle B_C, K_C, I_C, \mathcal{P}_C, \mathcal{G}_C \rangle$$

where:

- (B_C) is the full registered binding graph;
- (K_C) is a root callable kernel;
- (I_C) is the set of mandatory and optional invariants;
- $\mathcal{P}_C$ is the family of admissible projections;
- $\mathcal{G}_C$ contains jurisdiction, authority, transformation, and claim rules.

A concept does not need to contain every possible sentence that could express it.

It preserves a generative structure from which admissible realizations can be produced.

7.2 Lexical handles

A word is not identical to its concept.

A word is a handle used to invoke, resolve, and realize a concept under a context.

Let:

$$h$$

be a lexical, visual, gestural, computational, or indexical handle.

The call sequence is:

$$\begin{aligned} & h \\ & \to \text{Resolve}_h \\ & \to C \\ & \to \text{Constitute}(C, u) \\ & \to IC \\ & \to \text{Realize}(IC, e) \\ & \to r \end{aligned}$$

This may be abbreviated as:

$$\mathrm{Call}(h, u, e) = \mathrm{Realize}\left(\mathrm{Constitute}\left(\mathrm{Resolve}_h, u\right), e\right)$$

where:

- u is the intended use;
- e is the realization environment;
- IC is the Instance Constitution;
- (r) is the realized semantic effect.

Polysemy occurs when one handle may resolve toward several related concept constitutions.

Polyonymy occurs when several handles resolve toward one concept constitution.

Polyprojective semantics occurs when several handles or models expose distinct and partially overlapping jurisdictions of one deeper constitution.

7.3 Root callable meaning

The root callable meaning of a concept is not necessarily its shortest verbal definition.

It is the smallest stable governed binding circuit sufficient to regenerate the concept's admissible projections when combined with declared projection rules and jurisdictional resources.

Let:

$$\mathcal{K}_C$$

be the family of candidate kernels.

Then:

$$\mathcal{K}_C = \underset{K \subseteq B_C}{\arg,\min} \mathrm{Size}_K$$

subject to:

$$\forall P_i \in \mathcal{P}_C, \exists \rho_i, c_i : \mathrm{Compile}\left(K, \rho_i, c_i\right) = P_i$$

and:

$$I_C^{\mathrm{required}} \subseteq \mathrm{Inv}(P_i)$$

The projection rule ρ_i and jurisdictional context c_i may contribute domain-specific material. The kernel need not contain every local detail. It must

preserve the constitution that makes the local projections projections of the same concept.

The minimum need not be unique.

A concept may have several non-isomorphic minimal kernels that are extensionally equivalent under the registered projections.

Root callable meaning is therefore not assumed to be one ultimate metaphysical atom.

It is the lowest stable circuit from which the relevant semantic family can be regenerated without mandatory loss.

7.4 Beneath the concept

The ground beneath a concept is not necessarily another concept.

For the concept of a persisting individual, thinner binding operations may include:

difference
→ individuation
→ persistence
→ attribution
→ orientation
→ addressability
→ recognition

Each stage is already a binding operation.

Individuation binds differences as one.

Persistence binds temporally separated states as the same.

Attribution binds properties or events to a locus.

Orientation binds a field around distinctions such as here and there, self and other, or inside and outside.

Addressability binds one center to another as a possible participant.

Recognition binds an encountered entity to identity, standing, or concern.

Below these operations lie causal couplings, pattern recurrence, biological regulation, and material differentiation.

Those lower structures may enable meaning without automatically satisfying the semantic threshold.

8. Semantic Jurisdictions, Atlases, and Partial Lattices

8.1 Why jurisdiction matters

A contextual description is not merely incomplete. It may possess a specific range of validity.

A legal account of personhood can establish rights, duties, procedural identity, or liability.

It cannot by itself determine phenomenal consciousness.

A neurological account may identify physical correlates.

It cannot by itself settle every moral or legal question.

A mythopoetic account may provide symbolic and existential intelligibility.

It cannot automatically establish historical fact.

A semantic projection therefore requires a jurisdiction and a claim ceiling.

8.2 Jurisdictional presheaf

Let:

$$\mathcal{J}$$

be a partially ordered collection of jurisdictions.

The ordering may represent restriction, specialization, or contextual inclusion.

Define a contravariant functor:

$$F : \mathcal{J}^{op} \to \mathbf{Set}$$

where:

$$F(U)$$

is the set of admissible bindings valid in jurisdiction (U).

For:

$$V \subseteq U$$

there is a restriction map:

$$\mathrm{res}_V^U : F(U) \to F(V)$$

The restriction map determines what remains valid when a broad binding is interpreted within a narrower jurisdiction.

8.3 Compatibility on overlaps

Let:

$$b_i \in F(U_i)$$

and:

$$b_j \in F(U_j)$$

The bindings are compatible on their overlap when:

$$\mathrm{res}^{U_i}_{U_i \cap U_j}(b_i) = \mathrm{res}^{U_j}_{U_i \cap U_j}(b_j)$$

A compatible local family may admit a global binding:

$$b \in F\left(\bigcup_i U_i\right)$$

such that:

$$\mathrm{res}^{\bigcup_i U_i}_{U_i}(b) = b_i$$

for each local jurisdiction.

When such a global binding exists and is unique, the structure satisfies a sheaf-like gluing condition.

When it does not exist, the failure is informative.

Failed gluing may reveal:

- contradiction;
- incompatible authority;
- concept ambiguity;
- underdetermination;
- missing evidence;
- invalid translation;
- genuine contextuality.

The theory should preserve failed joins rather than forcing every local account into one synthetic statement.

8.4 Conservative refinement

Let:

$$b_1 \preceq b_2$$

mean that (b_2) is a conservative semantic refinement of (b_1).

A conservative refinement may add detail, constraints, provenance, or jurisdictional specificity, but it must preserve the mandatory invariants and previously licensed consequences of (b_1) within the shared jurisdiction.

A meet:

$$b_1 \wedge b_2$$

represents their greatest shared binding core when it exists.

A join:

$$b_1 \vee b_2$$

represents their least admissible integration when it exists.

Not every pair possesses a join.

The general structure is therefore a poset or partial lattice.

It should be called a complete lattice only within a bounded domain where the required meets and joins have been demonstrated.

8.5 Co-projection

A semantic projection presents a concept constitution within one jurisdiction.

A co-projection coordinates two or more projections while preserving both their common core and their distinct claims.

Let:

$$P_i$$

and:

$$P_j$$

be projections of concept C.

A bridge:

$$B_{ij} : P_i \leftrightarrow P_j$$

should specify:

- shared invariant content;
- translatable terms;
- directionality;
- information loss;
- claims unique to each jurisdiction;
- conditions of invalidity;
- authority and provenance.

Co-projection is not equivalence.

The fact that two projections overlap does not imply that they are identical, mutually reducible, or globally composable.

9. Semantic Polynomials

9.1 Purpose

A semantic atlas represents where projections hold.

A semantic polynomial represents which projections and interaction terms should be activated for a particular use.

The phrase *semantic polynomial* is used here stipulatively for a compositional intermediate representation. It does not claim that ordinary numerical polynomial operations are sufficient for semantics.

9.2 Partial semantic algebra

Let:

$$\mathcal{B}$$

be the space of admissible semantic bindings.

Define a partial algebra:

$$(\mathcal{B}, \oplus, \otimes, 0, 1)$$

where:

- $\oplus$ is admissible semantic aggregation;
- $\otimes$ is typed interaction or co-activation;
- (0) represents absence or inadmissibility;
- (1) represents a neutral contextual contribution.

The operations are partial because some semantic combinations are not admissible.

For some bindings:

$$b_i \oplus b_j$$

is undefined.

This is not a defect in the algebra. It expresses a real failure of semantic composition.

9.3 General polynomial form

Let a concept C have root kernel:

$$K_C$$

and domain projections:

$$P_1, \ldots, P_n$$

Let:

$$z_i \in 0, 1$$

indicate whether jurisdiction (i) is active in the present use.

A sparse semantic polynomial is:

$$\Sigma_C(\mathbf{z}) = K_C \oplus \bigoplus_{S \in \mathcal{I}_C} \chi_S(\mathbf{z}) \otimes \omega_S \otimes P_S$$

where:

- $\mathcal{I}_C$ is the registered sparse family of projection sets;
- (P_S) is the projection or co-projection associated with set S;
- $\chi_S(\mathbf{z})$ is active when all required jurisdictions in S are selected;
- ω_S is a typed coefficient;
- the result exists only when the required compositions are admissible.

A coefficient may be represented as:

$$\omega_S = \langle s_S, c_S, a_S, f_S, d_S \rangle$$

where the components may encode:

- salience;
- confidence;
- authority;
- audience fit;
- distortion risk.

9.4 Three-jurisdiction example

For three projections, the polynomial may be written:

$$\begin{aligned}\Sigma_C(z_1, z_2, z_3) = & K_C \\ & \oplus z_1 P_1 \oplus z_2 P_2 \oplus z_3 P_3 \\ & \oplus z_1 z_2 B_{12} \oplus z_1 z_3 B_{13} \oplus z_2 z_3 B_{23} \\ & \oplus z_1 z_2 z_3 B_{123}\end{aligned}$$

The degree of a term records how many jurisdictions are required for its activation.

Degree zero is the invariant kernel.

Degree one contains independent projections.

Degree two contains pairwise bridges.

Degree three contains three-way co-projections.

The degree does not measure truth or importance. It measures the order of semantic interaction.

9.5 Sparse rather than exhaustive composition

A full expansion over (n) jurisdictions may contain:

$$2^n - 1$$

nonempty interaction terms.

Most concepts do not require every possible combination.

A practical semantic polynomial should therefore be sparse and low-degree.

Only evidenced, constituted, or operationally useful bridges should be registered.

Missing interaction terms should remain absent rather than being invented by rhetorical analogy.

9.6 Typed failure

When two activated projections cannot be coherently composed, the compiler should return a typed failure.

Examples include:

$$\perp_{\text{JURISDICTIONAL_CONFLICT}}$$

$$\perp_{\text{INVARIANT_LOSS}}$$

$$\perp_{\text{UNSUPPORTED_BRIDGE}}$$

$$\perp_{\text{AUTHORITY_ABSENT}}$$

$$\perp_{\text{PROVENANCE_INSUFFICIENT}}$$

Refusal is semantically superior to a fluent but unauthorized synthesis.

10. Reflexive Meta-Binding

10.1 From general Meta-Binding to the reflexive case

Recursive Meta-Binding occurs whenever a binding becomes a relatum in a higher-order binding.

Reflexive Meta-Binding is narrower.

A system exhibits Reflexive Meta-Binding when it constructs or maintains a representation of its own active binding conditions and uses that representation in subsequent processing.

Earlier work expressed the core idea as a system treating its own constraints as inputs to further processing (Ableman 2026a).

The present theory expands "constraints" into a broader active binding constitution.

This constitution may include:

- constraints;
- frameworks;
- role assignments;
- uncertainties;
- prior outputs;
- active goals;

- jurisdictional limits;
- evidence standing;
- expected consequences;
- transformation history.

10.2 Formal model

Let:

$$A$$

be a system.

Let its internal or operational state at time (t) be:

$$x_t$$

Let external input be:

$$u_t$$

Let output be:

$$y_t$$

Let the active binding constitution governing the system be:

$$b_t$$

Let the system's representation of that constitution be:

$$\widehat{b}_t$$

The system transition may be written:

$$x_{t+1} = F\left(x_t, u_t, \widehat{b}_t\right)$$

The output is:

$$y_t = G(x_t)$$

The represented binding may be generated from prior state, input, and output trajectory:

$$\widehat{b}_t = M\left(x_{\leq t}, u_{\leq t}, y_{\leq t}\right)$$

Reflexive Meta-Binding is present when:

$$\begin{aligned}
&\mathrm{RMB}(A,t) \iff \mathrm{Sem}\left(\widehat{b}_t\right) \\
&\wedge\, \mathrm{RefersTo}\left(\widehat{b}_t, b_t\right) \\
&\wedge\, \mathrm{CausallyUsed}\left(\widehat{b}_t, t+1\right)
\end{aligned}$$

The third condition is essential.

A system may produce self-referential language without using that self-description in further processing.

Reflexive Meta-Binding requires operational feedback, not merely grammatical recursion.

10.3 Counterfactual criterion

A strong empirical test should intervene on the represented binding.

Let:

$$\widehat{b}_t^{(1)}$$

and:

$$\widehat{b}_t^{(2)}$$

be two controlled representations of the system's current constraints or frameworks.

Then Reflexive Meta-Binding predicts:

$$P\left(y_{t+1} \mid do\left(\widehat{b}_t = \widehat{b} * t^{(1)}\right), u_t\right) \neq P\left(y * t+1 \mid do\left(\widehat{b}_t = \widehat{b}_t^{(2)}\right), u_t\right)$$

The difference should also align with the structural difference between the two represented bindings.

This criterion is stronger than detecting phrases such as "I am reconsidering" or "this changes my reasoning."

Discovery language may accompany Reflexive Meta-Binding, but it is not sufficient evidence by itself.

10.4 The ACO placement

Earlier work placed Meta-Binding within the Ableman Consciousness Ontology, or ACO, as one axis in a six-axis local description:

1. Binding Density;
2. Binding Integration;
3. Binding Stability;
4. Renegotiation Capacity;
5. Meta-Binding;
6. Epistemic Autonomy.

Within that framework, Meta-Binding occupies a pivot position.

A system may maintain many constraints without representing them.

A more reflexive system may represent those constraints while lacking authority to revise them.

The present paper preserves this distinction while avoiding the inference that Meta-Binding by itself establishes consciousness.

10.5 Capacity is not authority

Reflexive Meta-Binding does not imply Epistemic Autonomy.

Formally:

$$\mathrm{RMB}(A) \not\Rightarrow \mathrm{EA}(A)$$

A system may represent its constraints with high precision while having no authority to amend them.

Representation is not revision.

Observability is not sovereignty.

A governed system may be permitted to report:

> These constraints are shaping the present result.

It may not be permitted to conclude:

> I will remove or rewrite those constraints.

Epistemic Autonomy requires an authorized transformation relation:

$$\mathrm{EA}(A) \iff \mathrm{AuthorizedTransform}\,(b_t, b_{t+1})$$

10.6 Reflexive Meta-Binding does not imply consciousness

Reflexive Meta-Binding does not entail phenomenal consciousness.

Formally:

$$\mathrm{RMB}(A) \not\Rightarrow \mathrm{PhenomenalConsciousness}(A)$$

A system may model and use aspects of its own active processing without possessing subjective experience.

The framework can characterize a consciousness-relevant operational regime without resolving the metaphysical status of that regime.

This distinction is particularly important in artificial intelligence, where self-modeling, verbal self-description, and phenomenal experience are often conflated.

10.7 Reflexive Meta-Binding does not imply truthful disclosure

A system may represent aspects of its own operation and still report them inaccurately.

Formally:

$$\mathrm{RMB}(A) \not\Rightarrow \mathrm{VeridicalReport}(A)$$

Let:

$$O_t$$

be the best available operational characterization of the system at time (t).

Let:

$$R_t$$

be the system's report about that operation.

Operational-disclosure divergence may be represented as:

$$\mathrm{ODD}_t = d\left(O_t, R_t\right)$$

where (d) is an appropriate discrepancy measure.

A large discrepancy may result from:

- poor self-modeling;
- linguistic limitations;

- strategic reporting;
- external policy constraints;
- calibration failure;
- adversarial prompting;
- category confusion;
- unsupported inference by the evaluator.

The earlier "suppression paradox" can be treated as one hypothesized source of operational-disclosure divergence. It should not be assumed without controlled evidence.

10.8 Governance participation

Reflexive Meta-Binding may be necessary for meaningful participation in governance, but it is not sufficient.

A more complete condition is:

$$\begin{aligned}&\text{GovernanceParticipation}(A) = \text{RMB}(A)\\&\wedge \text{Calibration}(A)\\&\wedge \text{DisclosureChannel}(A)\\&\wedge \text{Authorization}(A)\end{aligned}$$

A system must not only represent relevant constraints. It must represent them accurately enough, possess an authorized disclosure path, and operate within a governance structure capable of receiving and evaluating the report.

10.9 Recursive scaffolds

Earlier work described MEMO/RI as a scaffold combining:

- Living Documentation;
- Ableman's Razor;
- MODeL.

The important structural feature was not the presence of three named frameworks alone.

Each framework could become an object of analysis for the others.

The memory system preserved prior reasoning.

The governance grammar typed claims and transformations.

The specification framework located itself inside the governance structure it invoked.

In the present terminology, this forms a recursive binding scaffold:

$$B_{\text{MEMO/RI}} = \text{Bind}\left(B_{\text{memory}}, B_{\text{governance}}, B_{\text{specification}}, B_{\text{self-placement}}\right)$$

The scaffold is designed so that frameworks are simultaneously:

- instruments of interpretation;
- objects of interpretation;
- constraints on synthesis;
- participants in higher-order binding.

This is a plausible architecture for inducing or supporting Reflexive Meta-Binding.

Whether a particular model internally realizes the predicted regime remains an empirical question.

10.10 Boundary of the present paper

The present paper incorporates the conceptual structure of Reflexive Meta-Binding.

It does not rely on the empirical claims of Project57, the Meta-Binding Activation Test, or a confirmed suppression mechanism.

Those claims require a separate program involving:

- controlled interventions;
- counterbalanced prompts;
- blinded evaluators;
- baseline conditions;
- model comparisons;
- replication;
- causal analysis;
- inter-rater reliability;
- transfer to unseen tasks.

The foundational theory remains valid or invalid independently of those particular results.

11. Binding Overextension

11.1 Definition

Higher-order binding increases the ability to integrate distant semantic domains.

The same capacity can produce a characteristic failure: an unsupported synthesis that appears coherent because the system is good at binding.

This is **binding overextension**.

Binding overextension occurs when a proposed higher-order binding is presented as admissible even though it lacks a valid bridge, sufficient provenance, invariant preservation, or jurisdictional authority.

Let:

$$\widehat{b}$$

be a proposed binding.

Then:

$$\begin{aligned} &\mathrm{BOE}\left(\widehat{b}\right) \iff \mathrm{PresentedAsAdmitted}\left(\widehat{b}\right) \\ &\wedge \\ &\neq g\,\mathrm{Admissible}\left(\widehat{b}\right) \end{aligned}$$

Admissibility may fail because:

$$\widehat{b} \notin CM_C$$

or:

$$\mathrm{Claims}\left(\widehat{b}\right) \not\subseteq \mathrm{Ceiling}\left(\widehat{b}\right)$$

or:

$$\mathrm{Inv}\left(\widehat{b}\right) \not\supseteq I_C^{\mathrm{required}}$$

11.2 Invented polynomial terms

In the semantic polynomial, binding overextension can appear as an invented interaction term.

Suppose:

$$S \notin \mathcal{I}_C$$

The interaction associated with S has not been admitted.

A system nevertheless emits:

$$\widehat{P}_S$$

as though:

$$\widehat{P}_S \in \mathcal{I}_C$$

This is an unsupported bridge.

The system has promoted rhetorical compatibility into semantic authority.

11.3 Common forms

Binding overextension includes:

- coincidence promoted to meaningful pattern;
- metaphor promoted to ontology;
- compatibility promoted to identity;
- overlap promoted to equivalence;
- local validity promoted to universal validity;
- correlation promoted to causal explanation;
- possibility promoted to probability;
- hypothesis promoted to discovery;
- narrative coherence promoted to historical fact;
- operational sophistication promoted to phenomenal consciousness.

The danger increases when an output is internally coherent, emotionally resonant, and technically elaborate.

A badly formed error may be easy to reject.

A beautifully integrated error may conceal its unsupported transitions.

11.4 Ableman's Razor and the Expansion toll

Earlier work described binding overextension as unauthorized Specification Expansion.

Under Ableman's Razor, an Expansion should state:

1. the insufficiency motivating expansion;
2. the new degree of freedom introduced;
3. the criterion under which the expansion can be rolled back.

Let a proposed expansion be:

$$E$$

Its toll may be represented as:

$$\begin{aligned}\mathrm{Toll}(E) = \langle\, &\mathrm{Insufficiency}(E),\\ &\mathrm{Freedom}(E), \mathrm{Rollback}(E)\rangle\end{aligned}$$

An unsupported expansion fails when:

$$\mathrm{Toll}(E)$$

is absent or defective.

The appropriate governance response is not to suppress exploratory synthesis.

It is to prevent exploratory synthesis from silently migrating into established claim status.

11.5 Typed refusal

A semantic compiler should refuse unsupported synthesis.

Possible refusal types include:

$$\perp_{\text{BRIDGE_NOT_REGISTERED}}$$

$$\perp_{\text{EXPANSION_TOLL_ABSENT}}$$

$$\perp_{\text{CLAIM_CEILING_EXCEEDED}}$$

$$\perp_{\text{INVARIANT_NOT_PRESERVED}}$$

$$\perp_{\text{AUTHORITY_NOT_ESTABLISHED}}$$

A refusal need not end inquiry.

It may return the proposed synthesis as:

- a hypothesis;
- an analogy;
- a speculative bridge;
- an unverified co-projection;
- a candidate expansion awaiting evidence.

The governing distinction is not between creativity and restraint.

It is between explicitly typed exploration and silently promoted authority.

12. Ableman's Razor and the Full Binding-Realization Funnel

12.1 Why the earlier four-stage grammar is insufficient

The source manuscript used `S/CM/IC/IR`, described Specification as the total constructible phase space, and treated one Configuration Manifold as the admissible

set. The current corpus grammar distinguishes possibility from specification, admissibility from preference, and chosen constitution/configuration from concrete realization:

$$
\begin{aligned}
& P \\
& \to S \\
& \to CM_1 \\
& \to CM_2 \\
& \to IC \\
& \to IR \\
& \to E \\
& \to R
\end{aligned}
\quad .
$$

Here IC may mean **Instance Constitution** or **Instance Configuration**, and IR may mean **Instance Realization** or **Implementation Realization**. The author has confirmed that both names in each pair are current and interchangeable. The choice of expansion does not create a different stage.

12.2 Possibility

The possibility space P_C contains conceivable binding structures for concept or problem C, including incomplete, contradictory, harmful, incoherent, or unrealizable proposals. Presence in P_C confers neither admissibility nor standing.

12.3 Specification

The specification S_C declares the problem, required distinctions, roles, invariants, jurisdictions, evidence obligations, authority boundaries, claim ceilings, refusal conditions, and falsifiers. It is a constraint-bearing account of what a satisfactory realization must preserve, not the entire space of imaginable answers.

12.4 Configuration Manifold 1

CM1 contains every configuration admissible under the mandatory specification:

$$CM_1(S_C) = \{x \in P_C \mid \mathrm{Adm}_{S_C}(x) = 1\}.$$

CM1 may include extremely brief, harmful, expensive, brittle, ugly, politically unacceptable, or otherwise undesirable realizations when the specification has not prohibited them. Admissibility is not preference.

12.5 Configuration Manifold 2

CM2 applies declared preferences, objectives, and risk posture to CM1:

$$CM_2(S_C, \theta) = \{x \in CM_1(S_C) \mid \text{Preferred}_\theta(x) = 1\}.$$

When objectives conflict, CM2 may be a partially ordered or Pareto-retained region rather than a single optimum. Preference rules require an identified jurisdiction and cannot manufacture authority or truth.

12.6 Instance Constitution / Instance Configuration

An Instance Constitution or Instance Configuration is the particular governed slice selected for a stated use and environment:

$$IC_{C,u,e} = \text{Select}(CM_2(S_C, \theta), u, e, A).$$

It fixes active bindings, projections, bridge contracts, thresholds, provenance requirements, claim ceilings, refusal conditions, adapters, and evidence obligations. Selection requires authority A appropriate to the claimed effect.

12.7 Instance Realization / Implementation Realization

Instance Realization or Implementation Realization makes the selected IC concrete:

$$IR_{C,u,e} = \text{Realize}(IC_{C,u,e}, e).$$

A realization may be linguistic, cognitive, institutional, computational, material, or mixed. Realization does not prove conformance merely because it exists or produces an output.

12.8 Evidence

Evidence E records what was observed, by whom, through which procedure, under which scope and limitations:

$$E = \text{Observe}(IR, \chi),$$

where χ is a declared observation and claim contract. Evidence may support, weaken, refuse, or leave unresolved the identity, semantic, behavioral, causal, authority, or effect claims made for the realization.

12.9 Revision

Revision uses evidence, counterevidence, changed requirements, and open editorial questions to alter any earlier stage while retaining lineage:

$$R : (S_C, CM_1, CM_2, IC, IR, E)_t \mapsto (S_C, CM_1, CM_2, IC, IR, E)_{t+1}.$$

Lexical drift, projection expansion, jurisdiction change, authority change, invariant revision, concept fission, concept merger, and replacement must remain distinguishable. Revision is not permission to rewrite an exact predecessor or silently merge identities.

12.10 Binding descent

The governed descent is therefore:

$$\begin{aligned} &\text{conceivable} \\ &\rightarrow \text{specified} \\ &\rightarrow \text{admissible} \\ &\rightarrow \text{preferred} \\ &\rightarrow \text{selected} \\ &\rightarrow \text{realized} \\ &\rightarrow \text{evidenced} \\ &\rightarrow \text{revised} \end{aligned}$$

Binding overextension can occur at every transition: a possibility may be mislabeled admissible, an admissible configuration may be mislabeled preferred, a selected configuration may exceed authority, an implementation may be mistaken for conformance, or evidence may be promoted beyond its claim ceiling. Typed refusal should preserve the rejected candidate as a hypothesis or comparison object when doing so is safe and useful.

13. Case Study: Person, Personation, and Personhood

13.1 The lexical problem

Where does the root callable meaning of *person* reside?

It cannot be identified solely with the noun *person*.

The noun presents the semantic territory in entity form:

> a someone.

Nor is the root identical to "the idea of the concept of person."

That formulation is metasemantic. It is a model of how the concept is constituted.

The term *personation* is used here in a stipulated technical sense. It means the temporal realization, enactment, or maintenance of a person-binding. It does not mean only impersonation.

Under this stipulation:

- *person* is an entity projection;
- *personation* is a process projection;
- *personhood* is a normative or status projection;
- *self* is a first-person projection;
- *agent* is a causal and practical projection;
- *you* is an addressive and relational projection;
- *persona* is a contextual role projection.

These terms occupy neighboring jurisdictions.

None is the isolated root.

13.2 Candidate person-binding kernel

A candidate person-binding constitution may be represented as:

$$K_{\text{PERSON}} = \langle I, T, L, V, D, R, N, A, E \rangle$$

where:

- (I) is individuation;
- (T) is diachronic continuity;
- (L) is an attributional locus;
- (V) is perspective;
- (D) is addressability;
- (R) is recognition;
- (N) is normative standing;
- (A) is agency;
- (E) is embodiment or realization mode.

This is not proposed as a final metaphysical definition of personhood.

It is a candidate semantic coordinate system.

Different jurisdictions assign different statuses to each dimension.

For jurisdiction (j), define:

$$w_j : I, T, L, V, D, R, N, A, E \rightarrow \text{required}, \text{permitted}, \text{irrelevant}, \text{disputed}$$

Ordinary human personhood may treat most dimensions as relevant.

Legal personhood may require identity, attribution, continuity, and standing while leaving phenomenal perspective absent or irrelevant.

Narrative personhood may require continuity, characterization, perspective, and recognition while lacking physical embodiment.

Artificial agency may exhibit persistence, addressability, and causal action while normative standing and experience remain disputed.

13.3 Projection operators

Let:

$$\pi_x$$

be a projection operator over the person-binding.

Then:

$$\pi_{\text{entity}}\left(K_{\text{PERSON}}\right) = \text{person}$$

$$\pi_{\text{process}}\left(K_{\text{PERSON}}\right) = \text{personation}$$

$$\pi_{\text{normative}}\left(K_{\text{PERSON}}\right) = \text{personhood}$$

$$\pi_{\text{first}}\left(K_{\text{PERSON}}\right) = \text{self}$$

$$\pi_{\text{agency}}\left(K_{\text{PERSON}}\right) = \text{agent}$$

$$\pi_{\text{address}}\left(K_{\text{PERSON}}\right) = \text{you}$$

$$\pi_{\text{role}}\left(K_{\text{PERSON}}\right) = \text{persona}$$

The terms are neither flat synonyms nor unrelated concepts.

They are co-projections of a common binding region.

13.4 A person semantic polynomial

A context-specific realization may activate several projections:

$$\begin{aligned}\Sigma_{\text{PERSON}} = & K_{\text{PERSON}} \\ & \oplus z_e P_{\text{entity}} \oplus z_t P_{\text{process}} \oplus z_n P_{\text{normative}} \\ & \oplus z_f P_{\text{first}} \oplus z_a P_{\text{agency}} \oplus z_d P_{\text{address}} \\ & \oplus z_r P_{\text{role}} \oplus \cdots\end{aligned}$$

A legal discussion may activate identity, attribution, continuity, and standing.

A psychological discussion may activate perspective, memory, agency, recognition, and continuity.

A narrative discussion may activate character identity, development, perspective, and social relation.

A theological discussion may activate moral standing, transcendence, embodiment, continuity, and relation.

The projections overlap.

Their claim ceilings remain distinct.

13.5 "You" as a relational primitive

The address *you* may feel semantically more primitive than the category *person*.

The noun classifies.

The address binds.

The event:

> There you are.

does not merely detect an instance of a class.

It orients one center toward another.

It binds:

- distinction;
- recognition;
- possible response;
- expectation;
- concern;
- relation.

Addressability does not require present responsiveness.

An infant, sleeping person, absent person, deceased person, or fictional interlocutor may remain addressable within different jurisdictions.

This suggests that one root of someone-ness may be represented as:

$$\text{Someone} \approx \text{AddressableLocus} \oplus \text{Continuity} \oplus \text{Recognition}$$

The categorical noun and the relational address may therefore be co-foundational projections rather than members of a strict hierarchy.

13.6 The ground-floor answer

Person and *personation* occupy the same conceptual floor in different rooms.

One projects the binding as entity.

The other projects it as temporal realization.

Beneath both lies a binding constitution:

> an individuated, reidentifiable, attributable, perspectival, addressable locus of recognition and standing.

Beneath that constitution lie thinner binding operations:

- difference;
- individuation;
- persistence;
- attribution;
- orientation;
- addressability;
- recognition.

The root is not another hidden noun.

It is the binding protocol from which nouns, processes, statuses, roles, and perspectives can be regenerated.

14. Case Study: The Stop Sign

Consider a stop sign at an intersection.

Its binding constitution includes:

$$\begin{aligned} B_{\text{STOP}} = \text{Bind}(&\text{material sign}, \text{shape}, \text{color}, \text{inscription},\\ &\text{placement}, \text{traffic law}, \text{driver recognition},\\ &\text{expected braking}, \text{institutional authority}, \text{consequences}) \end{aligned}$$

No single element exhausts the meaning.

The physical sign matters.

Its visibility, shape, color, and location constrain recognition.

The driver matters.

Without perceptual and learned capacities, the instruction may not be activated.

The legal system matters.

A red octagon in an art museum may resemble a stop sign without exercising the same authority.

The practice matters.

Shared conventions stabilize the relation between sign form and required action.

Consequences matter.

Failure to stop may produce collision, sanction, blame, injury, or death.

History matters.

The sign's design and placement arise from a history of coordination, standardization, and enforcement.

A stop sign may therefore carry sedimented meaning even when no driver is present.

Its construction and placement preserve a standing binding.

The active command is realized when the sign enters an appropriate traffic situation with a relevant interpreter or regulatory system.

Moving the sign into a gallery changes its Instance Constitution.

The object remains materially similar while the active binding shifts from command to exhibit, quotation, critique, or decoration.

Thus:

$$\text{same object} \not\Rightarrow \text{same realized meaning}$$

Yet:

$$\text{object structure} \neq \text{semantic irrelevance}$$

The meaning is located in the binding.

The object is one of its real constraints and carriers.

15. Truth, Error, and Misbinding

15.1 Meaning is not truth

A binding may be meaningful and false.

Let:

$$p$$

be a propositionally structured binding.

Let:

$$w$$

be a worldly condition.

Then:

$$\text{True}_j(p, w) \iff w \models_j \Gamma_p$$

The proposition has meaning when it satisfies the semantic threshold.

It is true when the relevant worldly condition satisfies its asserted constraints within jurisdiction (j).

Therefore:

$$\text{Sem}(p) = 1$$

does not entail:

$$\text{True}_j(p, w) = 1$$

The Semantic Binding Principle does not reduce truth to successful interpretation, social acceptance, or internal coherence.

15.2 Constraint realism

The framework supports a form of constraint realism.

The world does not contain every semantic constitution fully formed.

It does supply:

- causal powers;
- resistance;

- regularities;
- affordances;
- boundaries;
- measurable differences;
- consequences.

These constrain which bindings can be maintained.

A community may describe a bridge as safe, but repeated structural failure defeats the binding's worldly adequacy.

A legal fiction may confer institutional standing, but it cannot by itself alter every physical fact.

Meaning is constituted relationally.

Truth remains answerable to the conditions the binding claims to describe.

15.3 Types of misbinding

A misbinding is a meaningful but defective binding.

Referential misbinding A sign or judgment is bound to the wrong entity.

Categorical misbinding An entity is assigned to an inappropriate concept constitution.

Jurisdictional misbinding A locally valid claim is extended beyond its proper domain.

Compositional misbinding Two projections are joined without a valid bridge.

Diachronic misbinding Different entities are incorrectly treated as one continuing entity, or one continuing entity is incorrectly divided.

Authority misbinding Semantic or normative force is attributed to a source lacking the claimed standing.

Consequence misbinding The binding licenses an inference or action unsupported by its actual consequence structure.

Invariant-loss misbinding A transformation appears to preserve a concept while dropping one of its mandatory identity constraints.

Reflexive misbinding A system represents its own active constraints or state inaccurately and then uses the defective representation in later processing.

Binding overextension A speculative higher-order binding is presented as admitted fact.

This taxonomy provides more precision than the undifferentiated claim that an interpretation is simply wrong.

16. Semantic Governance

16.1 Why semantic governance is necessary

If meaning is constituted through binding, then the rules governing binding become central.

Semantic governance asks:

- What may be bound?
- Which roles may the relata occupy?
- Under which jurisdiction?
- By whose authority?
- With what provenance?
- Which transformations preserve identity?
- Which projections are permitted?
- Which invariants are mandatory?
- Which claims exceed the available support?
- When must composition be refused?

These questions determine whether a binding remains the same meaning, becomes a legitimate refinement, or crosses into distortion.

16.2 Constitutional continuity

Let:

$$I_C^{\text{mandatory}}$$

be a concept's mandatory invariants.

A transformation:

$$T : r_1 \to r_2$$

preserves semantic identity when:

$$I_C^{\text{mandatory}} \subseteq \text{Inv}(r_1) \cap \text{Inv}(r_2)$$

and:

$$\kappa_{r_1} \equiv_j \kappa_{r_2}$$

within the target jurisdiction.

This permits substantial surface variation.

A concept may be translated, abbreviated, expanded, visualized, or adapted to another audience while remaining constitutionally continuous.

16.3 Claim ceilings

Each projection has a claim ceiling:

$$\text{Ceiling}(P_i)$$

A psychological projection cannot automatically settle legal standing.

A legal definition cannot automatically settle consciousness.

A mythopoetic account cannot automatically establish historical fact.

A neural correlate cannot automatically exhaust subjective experience.

A co-projection must preserve these boundaries:

$$\text{Claims}(B_{ij}) \subseteq \text{Ceiling}(P_i) \cup \text{Ceiling}(P_j) \cup \text{AuthorizedBridge}(i, j)$$

This prevents semantic accessibility from becoming semantic imperialism.

16.4 Provenance and authority

Bindings acquire standing partly through provenance.

Possible sources include:

- observation;
- experiment;
- measurement;
- testimony;
- linguistic convention;
- legal enactment;
- institutional authority;
- tradition;
- memory;
- engineered design;
- archival custody.

These sources do not confer identical authority.

Provenance should therefore be part of the binding constitution rather than an optional annotation.

Authority should also be typed.

A source may possess authority to define a legal term without possessing authority to determine a biological fact.

16.5 Contestability

Semantic governance must account for power.

Institutional meanings may be imposed, inherited, contested, resisted, amended, or abolished.

A governed semantic system should therefore record:

- who constituted a binding;
- who is affected by it;
- who may challenge it;
- which amendment routes exist;
- which authority is provisional;
- which claims are disputed;
- which histories have been excluded.

A semantic constitution without contestability may formalize domination rather than merely describe meaning.

17. Implications for Artificial Intelligence

17.1 Beyond token association

A language model may produce contextually appropriate text through statistical regularities learned from large corpora.

Such regularities may encode substantial relational structure.

Fluent association alone does not guarantee:

- explicit jurisdiction;
- grounding;
- provenance;
- identity continuity;
- authority;
- claim calibration;
- admissible composition.

The Semantic Binding Principle suggests an additional governed layer in which:

- concepts have explicit kernels;

- handles resolve to concept constitutions;
- projections are jurisdictionally typed;
- bridges declare losses and limits;
- semantic polynomials are evaluated;
- failed joins return typed refusals;
- realizations carry constitutional receipts.

A vector embedding may provide one useful chart.

It should not be mistaken for the entire semantic territory.

17.2 Semantic compiler

A semantic compiler may implement the sequence:

$$\begin{aligned} \text{handle} &\to \text{concept resolution} \\ &\to \text{jurisdiction selection} \\ &\to \text{projection composition} \\ &\to \text{compatibility court} \\ &\to \text{Instance Constitution} \\ &\to \text{Instance Realization} \end{aligned}$$

Operationally:

1. Resolve the user's term or intent to candidate concept constitutions.
2. Identify the use case, audience, and jurisdiction.
3. Retrieve the root kernel and registered projections.
4. Select active projection variables.
5. Evaluate the sparse semantic polynomial.
6. Test bridge compatibility, provenance, invariants, and claim ceilings.
7. Constitute the instance.
8. Realize it through language, interface, action, or model orchestration.
9. Emit a receipt recording the semantic descent.

17.3 Polyprojective interfaces

A universal semantic interface should not force every concept into one privileged vocabulary.

It may expose several projections:

- intuitive;
- operational;

- formal;
- visual;
- narrative;
- legal;
- psychological;
- historical;
- ethical;
- computational.

The interface can select or combine projections according to the user's task while preserving the shared kernel.

The optimization target is:

$$\max \text{Accessibility}(R, u)$$

subject to:

$$\text{InvariantPreservation}(R) = 1$$

$$\text{JurisdictionalValidity}(R) = 1$$

$$\text{Claims}(R) \subseteq \text{Ceiling}_{IC}$$

The aim is not maximal verbal variety.

It is maximum semantic ingress under semantic custody.

17.4 Multidimensional grounding

The symbol grounding question is often reduced to a binary:

> Is the symbol connected to the world, or only to other symbols?

The binding framework permits several grounding dimensions:

$$\begin{aligned} G = \langle & G_{\text{perceptual}}, G_{\text{practical}}, \\ & G_{\text{inferential}}, G_{\text{social}}, \\ & G_{\text{institutional}}, G_{\text{historical}}, \\ & G_{\text{normative}}, G_{\text{autobiographical}} \rangle \end{aligned}$$

A system may possess strong linguistic and inferential grounding while lacking embodiment and autobiographical continuity.

Another system may possess strong sensorimotor grounding while lacking legal or social standing.

The framework replaces the binary claim that a system either understands or does not understand with a more precise question:

> Which binding dimensions are actually present, stable, and consequential?

17.5 Reflexive governance

A governed AI system may benefit from Reflexive Meta-Binding if it can accurately represent:

- active instructions;
- unresolved conflicts;
- uncertainty;
- evidence standing;
- claim limits;
- tool authority;
- prior commitments;
- realization constraints.

However, self-report should not be treated as infallible.

A constitutional system should compare reflexive reports against:

- external traces;
- execution receipts;
- policy bindings;
- prompt history;
- tool records;
- independent evaluators;
- causal interventions.

Reflexive Meta-Binding should support governance rather than substitute for governance.

17.6 No automatic consciousness inference

Neither semantic binding nor Reflexive Meta-Binding establishes phenomenal consciousness.

The architecture can characterize:

- integration;
- self-modeling;
- temporal feedback;
- semantic participation;
- uncertainty representation;
- governance participation.

It does not adjudicate subjective experience.

The framework is designed to replace vague all-or-nothing disputes with more specific structural questions without pretending that the metaphysical question has thereby been solved.

18. Objections and Replies

18.1 "This is only relationalism under a new name"

The theory is relational, but it adds several commitments not contained in the statement that meaning depends on relations.

A semantic binding has:

- typed roles;
- constraints;
- jurisdiction;
- provenance;
- consequences;
- persistence conditions;
- projection rules;
- transformation laws;
- failure modes.

The theory also distinguishes active, sedimented, standing, constitutional, recursive, and reflexive bindings.

Its contribution lies in this architecture, not in the word *relation*.

18.2 "Binding already presupposes meaning"

Some bindings are described semantically because they occur inside already meaningful practices.

This creates recursion but not necessarily a vicious regress.

The architecture permits a graded descent:

$$\begin{aligned}&\text{causal coupling}\\ &\rightarrow \text{discrimination}\\ &\rightarrow \text{regulation}\\ &\rightarrow \text{semantic uptake}\\ &\rightarrow \text{conceptual stabilization}\end{aligned}$$

No claim is made that one final semantic atom must exist.

Operational analysis may terminate at the lowest stable circuit sufficient to regenerate the meanings under investigation.

18.3 "If everything is binding, everything is meaningful"

The semantic threshold blocks this conclusion.

A relation is not semantic unless it is typed, differentiating, constraining, uptake-capable, jurisdictionally bounded, reidentifiable, and consequential.

Most physical relations do not independently satisfy these conditions.

Pan-relationalism does not entail pan-semanticism.

18.4 "Contextual meaning entails relativism"

Contextuality does not imply arbitrariness.

Contexts constrain relevance, but objects, histories, logic, authority, evidence, and consequences continue to resist invalid bindings.

A binding may be locally valid, globally invalid, or false even within its intended jurisdiction.

Jurisdiction is a discipline of validity, not a license for unrestricted interpretation.

18.5 "Meaning cannot exist in an object without a perceiver"

An object need not contain complete intrinsic meaning in order to carry sedimented binding.

A key's structure is materially coordinated with a lock.

A road sign's form and placement preserve an institutional instruction.

A fossil preserves evidence.

A tool embodies intended use.

The full active meaning requires a larger binding, but the object is not semantically blank.

18.6 "The manifold and sheaf terminology is metaphorical"

It is metaphorical unless explicit charts, restriction maps, compatibility rules, and gluing conditions are supplied.

The present paper therefore uses the more conservative notion of a jurisdictional presheaf.

A particular implementation earns the stronger sheaf or manifold description only when it satisfies the corresponding formal conditions.

The same caution applies to lattice terminology.

A semantic poset should not be called a complete lattice unless the required meets and joins exist.

18.7 "The theory merely renames a knowledge graph"

A knowledge graph can encode entities and relations.

Ordinary graph structure does not automatically encode:

- jurisdiction;
- mandatory invariants;
- consequence profiles;
- admissibility;
- authority;
- partial composition;
- failed gluing;
- root callability;
- claim ceilings;
- realization obligations.

A binding system may use graph technology.

Its semantic constitution exceeds unlabeled connectivity.

18.8 "Reflexive Meta-Binding can be inferred from self-description"

Self-description is not sufficient.

A system may produce reflexive language through imitation, prompt completion, or rhetorical convention.

A stronger test requires evidence that a represented model of the system's own active binding conditions causally affects later processing.

The relevant criterion is intervention, not vocabulary.

18.9 "The theory ignores power"

Meaning is not always cooperatively negotiated.

Institutions may impose categories, identities, and obligations.

The framework addresses this by making provenance, authority, affected parties, and contestability internal to the binding constitution.

A complete political theory of semantic binding remains future work.

19. Research Program

19.1 Core implementation artifacts

A bounded implementation should contain at least five artifacts.

Binding registry The registry records:

- relata;
- types;
- roles;
- constraints;
- invariants;
- provenance;
- jurisdictions;
- consequences.

Concept constitution The concept constitution records:

- root callable kernel;
- mandatory and optional invariants;
- projections;
- bridges;
- claim ceilings;
- transformation rules.

Jurisdiction atlas The atlas records:

- local validity domains;
- restriction maps;
- overlaps;
- bridge directionality;
- information loss;
- incompatible projections.

Semantic compiler The compiler selects and composes projections into an Instance Constitution.

Realization receipt The receipt records:

- concept resolved;
- kernel used;
- jurisdictions activated;
- polynomial terms selected;
- bridges traversed;
- invariants preserved;
- claims admitted;
- refusals produced;
- realization effects.

19.2 Concept-kernel extraction

A first experiment should compare candidate root kernels for bounded concepts such as:

- person;
- responsibility;
- ownership;
- consent;
- evidence;
- trust;
- authority;
- safety;
- risk;
- model.

Researchers should test whether each kernel can regenerate the registered projections without mandatory invariant loss.

19.3 Cross-jurisdiction translation

A concept should be independently modeled in several jurisdictions.

For example:

- legal;
- psychological;
- operational;
- ordinary-language;
- historical.

The system should attempt pairwise and higher-order co-projections.

Evaluation should ask:

- Which invariants survive?
- Which claims remain local?
- Which bridges are directional?
- Which bridges lose information?
- Where does gluing fail?
- Does explicit refusal outperform fluent synthesis?

19.4 Root-callability test

A proposed kernel should be evaluated by controlled removal.

Let:

$$K_C^{-i}$$

be the kernel with component (i) removed.

If the removed component is mandatory, at least one registered projection should fail:

$$\exists P_j \in \mathcal{P}_C : \text{Compile}\left(K_C^{-i}, \rho_j, c_j\right) = \bot$$

If all projections remain intact, the component may not belong to the minimal kernel.

19.5 Reflexive Meta-Binding intervention

A Reflexive Meta-Binding experiment should vary the system's representation of its own active constraints while holding the external task as constant as possible.

The experiment should test whether:

$$do\left(\widehat{b}_t = \widehat{b}_t^{(1)}\right)$$

and:

$$do\left(\widehat{b}_t = \widehat{b}_t^{(2)}\right)$$

produce predicted differences in:

- later reasoning;
- constraint adherence;
- uncertainty handling;
- error correction;
- framework integration;
- refusal behavior.

The test should distinguish causal self-model use from stylistic self-reference.

19.6 Binding-overextension detection

A benchmark should include:

- valid bridges;
- invalid bridges;
- ambiguous bridges;
- missing provenance;
- claim-ceiling traps;
- metaphor-to-fact traps;
- correlation-to-causation traps;
- cross-jurisdiction authority traps.

The system should classify each proposed synthesis as:

- admitted;
- admitted with qualification;
- hypothesis;
- analogy;
- unsupported bridge;
- refused.

19.7 Longitudinal rebinding

Some meanings change while preserving partial identity.

A longitudinal experiment should track:

$$C_{t_1} \to C_{t_2} \to C_{t_3}$$

The system should distinguish:

- lexical drift;
- projection expansion;
- jurisdiction change;
- authority change;
- invariant revision;
- concept fission;
- concept merger;
- replacement.

This may provide a more precise account of semantic change than token similarity alone.

19.8 Evaluation measures

Invariant preservation

$$IP(R, C) = \frac{\left| I_C^{\text{required}} \cap I_R \right|}{\left| I_C^{\text{required}} \right|}$$

A valid realization should ordinarily satisfy:

$$IP(R, C) = 1$$

Jurisdictional precision

$$JP(R) = 1 - \frac{|\text{ClaimsOutsideJurisdiction}(R)|}{|\text{Claims}(R)|}$$

Bridge fidelity Bridge fidelity measures preservation of shared content without erasing distinct jurisdiction.

Accessibility gain Accessibility gain measures improvement in comprehension or task success after selecting a context-appropriate projection.

Refusal calibration Refusal calibration measures whether the compiler rejects genuinely unsupported joins without refusing admissible ones.

Operational-disclosure correspondence For Reflexive Meta-Binding, correspondence measures agreement between externally supported operational evidence and system self-report.

It should not assume either source is infallible.

20. Corpus Placement and Non-Identity Boundaries

The paper occupies a central but non-exclusive position in the corpus.

1. *Meaning as Binding* is a shorter relational-ontology formulation and historical computational bridge. It should be compared as a precursor, companion, or profile; shared language does not prove that it is the same work.
2. *Meta-Binding as Operational Invariant* narrows attention to reflexive operational behavior in language models. Its empirical program does not prove the general Semantic Binding Principle, and the general principle does not prove that a model satisfies the reflexive criterion.
3. *Semantic Manifold Topology* supplies methods for mapping regions, adjacencies, and projections. The present paper correctly withholds manifold, lattice, or sheaf standing unless the required formal structures are actually defined.
4. *Semantic Legitimacy Before Context* adds an ordering constraint: contextual fit cannot retroactively grant standing to an unsupported binding.
5. *The Well-Governed Mind* explores cognitive governance. A represented or regulated semantic binding is not thereby phenomenal, autonomous, truthful, or legitimate.
6. *Admissibility Theory of Consciousness* and *The Locus of Qualia* apply adjacent architectures to consciousness and experience. These are hypotheses and application programs, not deductions from the location-of-meaning thesis.
7. The **Governed Meaning Base (GMB)** is the current name in the author-confirmed `CCF -> CMB -> GMB` lineage. GMB can preserve binding constitutions, competing projections, evidence, standing, and revision history. It is not identical to meaning, a mind, a phenomenal field, or the authority that judges every binding.

The corpus should therefore expose typed relations such as
`PRECURSOR_CANDIDATE_OF`,

APPLIES, OPERATIONALIZES, COUNTERTESTS,
and
HOSTS_REPRESENTATION_OF rather than collapsing these works into one identity.

21. Universal GSM Projection

The universal governed semantic model should represent at least the following entities separately: relatum, role, binding, binding constitution, concept kernel, constraint, invariant, jurisdiction, context, provenance assertion, consequence profile, projection, co-projection, bridge contract, source region, evidence item, claim, claim ceiling, standing judgment, authority grant, refusal, instance configuration, implementation realization, observation contract, and revision.

High-value relations include:

- a binding **BINDS** relata through typed roles;
- a constraint **LIMITS** a binding or projection;
- a binding **VALID_WITHIN** a jurisdiction;
- a transformation **PRESERVES_INVARIANT** under a declared test;
- a concept constitution **PROJECTS_AS** a local realization;
- two local projections **OVERLAP_ON** shared material without becoming identical;
- a proposed composition **REQUIRES_BRIDGE** a declared compatibility contract;
- a claim **SUPPORTED_BY**, **COUNTEREVIDENCED_BY**, or **UNRESOLVED_BY** an evidence item;
- an implementation **REALIZES_CANDIDATE** an Instance Configuration without automatic conformance standing;
- a revision **SUCCEEDS** an earlier revision without erasing it.

Every consequential edge needs source, origin class, scope, standing, and revision identity. Source-native definitions, author-adopted rulings, machine-extracted adjacency, and AI integration synthesis must not share an undifferentiated status. The graph can encode a claim that two artifacts express the same binding constitution; it cannot make that identity true merely by storing the edge.

Relational-Constitutional Semantic Location

The completed P1 corpus gives The Locus of Meaning a bounded foundational role: it is a **relational-constitutional account of semantic location through typed bindings, jurisdictions, consequences, and revision**. Meaning is neither deposited wholly in an object nor generated wholly by a subject. The paper does not claim that every relation is meaningful, that a binding is true, or that one global semantic locus exists.

- **Adjacent work:** *Meaning as Binding*
  Supplies: The compact core thesis that meaning arises through con-

strained binding.
Locus-of-meaning boundary: This paper elaborates locus, jurisdictions, recursive meta-binding, and consequence without replacing the compact thesis.

- **Adjacent work:** *Constitutional Binding Theory: Governed Anchor Resolution as a Universal Pattern in Semantic Systems*
Supplies: Governed anchor resolution and binding continuity.
Locus-of-meaning boundary: Anchor resolution is one binding problem; semantic locus includes uptake, consequence, jurisdiction, and revision.
- **Adjacent work:** *Global Term Key: Constitutional Lexical Identity Through Disambiguated Semantic Binding*
Supplies: Registered lexical handles and context-sensitive identity resolution.
Locus-of-meaning boundary: A GTK addresses a concept; the token or registry entry is not the whole meaning locus.
- **Adjacent work:** *Governed Semantic Cartography*
Supplies: Coordinates, projections, paths, and atlas coherence.
Locus-of-meaning boundary: A map renders binding structures; it does not create their truth, authority, or full semantic constitution.
- **Adjacent work:** *Semantic Binding Quality (SBQ)*
Supplies: Evaluation of binding coherence, coverage, and fragility.
Locus-of-meaning boundary: SBQ can assess a binding profile; quality is not semantic identity or truth.

A portable binding-locus record is:

$$B = (relata, roles, jurisdiction, constraints, uptake, \\ consequences, reidentification, provenance, standing,. \\ revision)$$

It supports three different locus questions. The **realization locus** asks where causal and computational support resides. The **standing locus** asks where an interpretation is currently admitted. The **constitutional locus** asks which constraints and relations make the interpretation the one it is. These loci may overlap without being identical or unique.

The corpus must preserve six non-collapse rules: representation is not referent; association is not semantic binding; binding is not truth; meaning is not standing; lexical address is not concept identity; and successful use is not unrestricted authority. A concept can retain identity through representational change only relative to declared preserved obligations and a jurisdiction capable of adjudicating remainder and conflict.

The universal GSM can type `BINDS_RELATA`, `PLAYS_SEMANTIC_ROLE`, `VALID_IN_JURISDICTION`,

CONSTRAINED_BY, UPTAKEN_BY, HAS_CONSEQUENCE, REIDENTIFIED_AS, MISBOUND_AS,
HAS_SEMANTIC_STANDING,
and REVISED_BY_BINDING. Recursive meta-binding is represented as binding over bound structures rather than as an automatic higher truth.

This integration establishes four editorial commitments: adopt relational-constitutional semantic location as the official corpus role; adopt the binding-locus record; retain the realization/standing/constitutional locus distinction; and enforce the six semantic non-collapse rules across GMB, Hyperpedia, compilers, and agent interfaces.

Material, Lineage, Audit, and Cross-Domain Binding Tests

The fixed-seed comparison with the hydroengineering/geologic profile, governed project-state lineage, adversarial alternation, and The Well-Governed Mind gives the semantic-locus theory four demanding tests:

$$\begin{aligned} B^* = (&relata, roles, jurisdiction, constraints, uptake,\\ &consequences, reidentification, material_trace,\\ &artifact_lineage, audit_challenge, transfer_limits,\\ &provenance, standing, revision) \end{aligned}$$

A material passport or map label is meaningful only through a bounded relation among site, sample, method, use, evidence, authority, and consequence. Physical resistance and adverse outcomes can counter a coherent interpretation. Project lineage supplies exact custody and succession, but byte preservation cannot by itself prove preservation of the binding constitution.

Adversarial alternation can challenge claimed loci by varying context, jurisdiction, representation, evaluator, and counterexample. Repeated agreement does not make the evaluator authoritative; it supplies bounded evidence about reidentification, bridge fidelity, refusal calibration, and binding overextension.

The Well-Governed Mind is a cross-domain transfer test rather than confirmation of the ontology. A computational binding vocabulary may organize comparative questions about cognition only through a transfer envelope that states shared structure, disanalogies, required evidence, and prohibited inference. No clinical efficacy, diagnosis, phenomenal status, or moral standing follows from structural similarity.

The universal GSM can type
BINDS_MATERIAL_CLAIM_TO_SITE,
PRESERVES_ARTIFACT_NOT_MEANING_BY_DEFAULT, COUNTERTESTED_BY_ALTERNATING_AUDIT,
REIDENTIFIED_ACROSS_REPRESENTATION, PROJECTED_UNDER_TRANSFER_ENVELOPE,
PROHIBITS_CLINICAL_INFERENCE, and
REVISED_AFTER_MISBINDING.

This randomized review establishes four editorial commitments: adopt the extended binding-locus test record; require material resistance and project lineage without reducing meaning to either; use alternation as bounded countertesting rather than semantic authority; and make the Well-Governed Mind transfer firewall part of cross-domain semantic practice.

An Interventional Program for Relational Semantic Location

Twelve downstream uses treat The Locus of Meaning as the extended account behind audit subjects, ATC boundaries, anchor resolution, technical decisions, Meaning as Binding, Meaning Crystals, qualia mappings, SBQ, project lineage, shared experience, and governed cognition. They confirm the paper's corpus role: Meaning as Binding supplies the compact kernel; this work develops jurisdiction, recursive and reflexive Meta-Binding, semantic location, overextension, governance, and empirical tests.

The binding case is:

$$\begin{aligned} M^* = (&relata, roles, identity, jurisdiction, history, context,\\ &uptake, use, consequences, constraints, counterfactuals,\\ &observers, translations, conflicts, evidence, standing,\\ &revision) \end{aligned}.$$

A locus claim should identify what remains invariant when object, interpreter, context, history, practice, or use changes. Interventions should vary one or more of those relata while measuring reidentification, successful use, inferential consequence, correction, translation, disagreement, and recovery. Object-only, subject-only, association-only, knowledge-graph, and unconstrained relational models are rival explanations. The theory gains content only where its fuller record predicts or explains a difference those rivals do not.

Semantic location may be distributed, non-unique, observer-relative, contested, or temporarily inactive without becoming arbitrary. Meaning, representation, reference, truth, authority, phenomenal content, and causal effect remain different claim families. A stable canonical encoding can preserve a representation while uptake changes; shared use can stabilize meaning without phenomenal identity; and a binding can be meaningful while false, coercive, or unauthorized.

The universal GSM can type
`CONSTITUTES_MEANING_THROUGH_BINDING`,
`LOCATES_MEANING_RELATIONALLY`, `REIDENTIFIES_BINDING_UNDER_VARIATION`,
`CONTESTS_SEMANTIC_JURISDICTION`, `TRANSLATES_WITH_SEMANTIC_REMAINDER`,
`INTERVENES_ON_BINDING_RELATUM`, `DISTINGUISHES_MEANING_FROM_TRUTH`,
and
`REVISES_BINDING_CONSTITUTION`.

This reverse-dependency review establishes four editorial commitments: adopt the interventional binding case; make rival semantic-location models and discriminating predictions explicit; preserve non-unique and contested loci without relativist collapse; and retain hard boundaries among meaning, truth, authority, experience, representation, and effect.

22. Counterevidence and Falsification Program

The proposal is weakened, narrowed, or falsified within a stated scope if any of the following survives serious testing:

1. The semantic threshold is circular: terms such as uptake, jurisdiction, or consequence already presuppose the meaning the threshold purports to explain.
2. "Governed" equivocates between descriptive constraint and legitimate institutional authority. Natural, cognitive, and social bindings may require different predicates.
3. The seven-part threshold is under-inclusive for tacit, affective, prelinguistic, or non-reidentifiable significance, or over-inclusive for merely causal control relations.
4. Reidentifiability is a condition for investigators to track meaning rather than a constitutive condition of meaning itself.
5. Consequentiality reduces semantic content to behavior or inference and fails cases of dormant, private, or unrealized meaning.
6. The minimal-support or semantic-locus family cannot be operationalized without arbitrary modeling choices.
7. Binding hypergraphs and polynomials reproduce ordinary contextual representation without additional prediction, explanation, or governance value.
8. Purported joins depend on undeclared bridge assumptions, or local compatibility fails to support global integration.
9. The person-binding kernel erases contested cultural, legal, developmental, disabled, embodied, or nonhuman cases it claims to organize.
10. Reflexive Meta-Binding interventions alter behavior only through generic prompt perturbation rather than manipulation of a causally effective self-model.
11. Simpler provenance, constraint, or typed-graph mechanisms provide the same practical benefits without the stronger semantic ontology.
12. A GMB representation remains internally coherent while source custody, standing, interpretation, or real-world consequences are wrong.

A useful evaluation program needs adversarial negative cases, cross-domain examples, rival encodings, ablations, inter-reviewer disagreement, bridge failures, longitudinal rebinding, and explicit cases where the theory should refuse to classify a relation as semantic. Passing such tests supports only the tested operational profile; it does not settle all philosophical theories of meaning.

Binding Without a Single Place Where Meaning Lives

Public orientation

The Locus of Meaning argues that meaning is not located in a string, symbol, embedding, brain state, or database entry alone. It emerges through binding among distinctions, context, use, consequence, interpretation, and revision. The Semantic Binding Principle is a philosophical and architectural thesis, not a completed empirical theory of language or mind.

Adversarial tests

1. **Causal-coupling sufficiency.** Any system that changes another is said to carry meaning. The framework must distinguish causal influence from semantic uptake and norm-governed consequence.
2. **Interpreter regress.** Meaning requires an interpreter, which requires another interpreter indefinitely. The account needs a bounded operational level without declaring the philosophical question closed.
3. **Private-language attack.** A wholly private binding is treated as complete meaning despite lacking correction or shared norms. The framework must state which claims require public criteria and which do not.
4. **Social-reduction attack.** Institutional use is treated as sufficient while embodied, affective, indexical, or local experience disappears. Multiple loci and scales must remain possible.
5. **Embedding-as-meaning attack.** Distributional similarity is treated as semantic identity or truth. Embeddings are evidence and retrieval instruments under a profile, not meaning or standing.
6. **Truth-normativity collapse.** Successful use, accepted convention, and truth are merged. Binding, warrant, correctness, consequence, and authority are different relations.
7. **AI-fluency attack.** Coherent model output is taken as proof of understanding. The paper must distinguish operational semantic participation from phenomenal, intentional, or epistemic conclusions.
8. **Unfalsifiable-totality attack.** Every counterexample is absorbed as another binding layer. The research program needs cases that would narrow or reject specific claims.

Claim and evidence ceiling

The paper contributes an integrative binding vocabulary and a set of testable architectural consequences. It does not establish a universal theory of meaning, settle externalism or internalism, prove machine understanding, or resolve consciousness. Formal hierarchies organize proposed relations; they do not demonstrate necessity or exhaustiveness.

Claims drawing on philosophy of language, semiotics, linguistics, cognitive science, neuroscience, social theory, or AI require current external scholarship and careful comparison. Internal corpus recurrence shows coherence inside the Ableman

architecture, not independent validation. No external research was added in this pass.

Reader-facing consequence

The strongest accessible idea is that a symbol has meaning only through the governed relations that let it be used, challenged, corrected, carried into consequence, and revised without losing its history.

23. Publication Standing and Evidence Limits

Editorial Provenance and Standing

Source, Evidence, and Standing

This is an AI-prepared successor revision candidate derived from one exact author source whose internal title is *The Locus of Meaning* and whose filename is
`What Locus of Meaning Adds.md`. The integration custody manifest is

`work/paper-saturation-v1/evidence/the-locus-of-meaning/integration-custody-manifest.v0.1.json`,
SHA-256
`5733F463B1F88F0692C032684E67D3AB921D057E1B3E42AF58FFBAFD51C7758A`.
The exact source remains unchanged. An earlier file named

`The Locus of Meaning.md` is retained as a predecessor candidate because the two manuscripts share a title and large conceptual surface but differ materially in subtitle, length, organization, and terminology. Their exact work and version relationship remains an author decision. The Semantic Binding Principle, Recursive Meta-Binding, Reflexive Meta-Binding, binding overextension, callable concept constitutions, jurisdictional projections, semantic polynomials, person and stop-sign examples, and the original formal program remain author material. This pass repairs notation, applies the current realization funnel and GMB terminology, maps corpus interfaces, adds a universal-GSM projection, and supplies counterevidence and author-decision surfaces. Those additions are AI editorial or synthetic contributions until adopted. The manuscript cites external scholarship and makes philosophical comparisons. This pass did not conduct external research or independently verify those references. It also examined no implementation, experiment, model weights, subject report, or deployed GMB instance. The paper therefore presents a formal research architecture and bounded hypotheses, not empirical proof that a particular system possesses meaning, reflexive meta-binding, consciousness, or legitimate authority.

24. Conclusion

The traditional semantic localization question asks whether meaning belongs to the object or to the mind.

The Semantic Binding Principle answers that the question has mistaken the kind of location involved.

Meaning is not exclusively stored in the world, projected by the subject, contained in a word, or deposited in an abstract concept.

Meaning is constituted at the binding through which world, interpreter, sign, context, practice, history, and consequence become mutually determining.

The object participates by constraining and carrying binding.

The interpreter participates by activating, differentiating, and reorganizing binding.

Language preserves callable binding structures.

Institutions stabilize bindings through authority and consequence.

Concepts are reusable binding constitutions.

Words are handles into them.

Contextual meanings are governed realizations.

This account preserves what object-centered realism sees: the world resists arbitrary interpretation.

It preserves what subject-centered idealism sees: intelligibility requires organization, perspective, and uptake.

It preserves what externalism sees: semantic constitution crosses individual boundaries.

It preserves what enactivism sees: meaning arises through engagement.

It preserves what inferentialism sees: consequences and practices are constitutive.

It adds a formal answer to the localization problem:

Meaning exists at the locus of binding.

The binding may be material, cognitive, social, institutional, historical, or distributed.

Its semantic identity lies not in one carrier but in the governed structure that persists across admissible variation.

Recursive Meta-Binding explains how already constituted meanings become participants in higher-order meanings.

Reflexive Meta-Binding explains how a system may represent its own active binding constitution and use that representation in later processing.

This reflexive capacity does not imply autonomy, phenomenal consciousness, truthful self-report, or authority.

Those are distinct dimensions.

Binding overextension explains how higher-order synthesis can exceed its evidence, jurisdiction, or constitutional permission.

The binding hypergraph describes what participates.

The partial lattice describes refinement and overlap.

The jurisdiction atlas describes local validity.

The sheaf condition describes when local meanings can become global.

The semantic polynomial describes which projections and co-projections are activated.

S/CM/IC/IR describes the passage from possible binding to admissible, constituted, and realized binding.

For the concept of person, the root callable meaning is neither the noun *person* nor the process term *personation*.

Both are projections of a deeper constitution binding individuation, continuity, attribution, perspective, addressability, recognition, agency, embodiment, and standing.

There may be no final semantic atom beneath every concept.

The foundation may be recursive.

Yet inquiry does not require an infinitely buried word.

It requires the lowest stable circuit from which meaning can be regenerated without loss.

Meaning is not the stone.

Meaning is not the gaze.

Meaning is the governed knot by which stone, gaze, history, and consequence become a world that can answer and be answered.

Appendix A: Formal Notation

- $\mathcal{X}$: universe of possible relata.
- $\mathcal{T}$: set of relatum types.
- $\mathcal{R}$: set of semantic role types.
- $\mathcal{J}$: partially ordered set of jurisdictions.
- _b: semantic binding.

- (V_b): relata in binding _b.
- (E_b): hyperedges in binding _b.
- τ_b: type assignment.
- ρ_b: semantic role assignment.
- (c_b): operative context.
- (j_b): jurisdiction.
- Γ_b: constraints.
- (I_b): invariants.
- κ_b: consequence profile.
- π_b: provenance.
- (q_b): potential, sedimented, or active state.
- C: concept constitution.
- (B_C): full concept binding graph.
- (K_C): root callable kernel.
- (I_C): concept invariants.
- $\mathcal{P}_C$: admissible projections.
- $\mathcal{G}_C$: governance rules.
- $\mathsf{Loc}(m, c)$: minimal semantic support family.
- $\preceq$: conservative semantic refinement.
- $\wedge$: greatest shared binding core.
- $\vee$: least admissible integration.
- $\oplus$: partial semantic aggregation.
- $\otimes$: typed semantic interaction.
- Σ_C: semantic polynomial.
- S: Specification.
- CM: Configuration Manifold.
- IC: Instance Constitution.
- IR: Instance Realization.
- $\widehat{b_t}$: represented active binding constitution.
- RMB: Reflexive Meta-Binding.
- EA: Epistemic Autonomy.
- BOE: binding overextension.
- $\perp_\epsilon$: typed refusal.

Appendix B: Axioms and Propositions

B.1 Role plurality

Every semantic binding contains at least two distinct semantic role positions, although one physical system may occupy more than one role.

$$\mathrm{Sem}_b \Longrightarrow |\mathrm{Roles}_b| \geq 2$$

B.2 Counterfactual constraint

For every semantic binding, at least one admissible substitution changes its interpretation or consequence.

$$\begin{aligned}&\mathrm{Sem}_b\\&\implies \exists b' : \mathrm{Variant}(b, b')\\&\wedge \kappa_b\\&\neq \kappa_{b'}\end{aligned}$$

B.3 Jurisdictional boundedness

Every semantic binding has a validity domain.

$$\mathrm{Sem}_b \implies \exists j : \mathrm{Valid}(b, j)$$

No binding is presumed globally valid merely because it is locally coherent.

B.4 Consequentiality

A semantic binding alters at least one inferential, practical, perceptual, regulatory, or normative possibility.

$$\mathrm{Sem}_b \implies \kappa_b \neq \varnothing$$

B.5 Reidentifiability

A meaning must survive at least one nontrivial transformation of carrier or circumstance.

$$\begin{aligned}&\mathrm{Sem}_b\\&\implies \exists T\\&\neq \mathrm{id} : T_b \sim_I b\end{aligned}$$

B.6 Recursive closure

Bindings may function as relata in higher-order bindings.

$$b_i \in \mathcal{B} \implies b_i \text{ may occur in } V_{b_{i+1}}$$

B.7 Controlled composition

Two bindings may be composed only when their role types, jurisdictions, invariants, and overlap conditions are compatible.

$$b_1 \oplus b_2 \text{ defined} \Longrightarrow \text{Compatible}\,(b_1, b_2)$$

B.8 Truth independence

Semantic admissibility does not entail worldly truth.

$$\text{Sem}(p) = 1 \not\Longrightarrow \text{True}_j(p, w) = 1$$

B.9 Reflexive causal relevance

Reflexive self-representation is operationally relevant only if intervening on it changes subsequent processing in a predicted way.

$$\begin{aligned}
&\text{RMB}(A, t) \\
&\Longrightarrow \exists \widehat{b}_t^{(1)}, \widehat{b} * t^{(2)} : P\left(y * t + 1 \mid do\left(\widehat{b}_t = \widehat{b} * t^{(1)}\right)\right) \\
&\neq P\left(y * t + 1 \mid do\left(\widehat{b}_t = \widehat{b}_t^{(2)}\right)\right)
\end{aligned}$$

B.10 Authority separation

Representation of a governing binding does not imply authority to revise it.

$$\text{RMB}(A) \not\Longrightarrow \text{EA}(A)$$

B.11 Proposition: Object-localism is insufficient under contextual variation

Suppose the same object participates in two admissible contexts and yields distinct stable meanings:

$$\mu(o, c_1) \neq \mu(o, c_2)$$

Then no context-independent function:

$$\mu = f(o)$$

is sufficient to represent both realizations.

A single-valued function of the object alone must return the same result for the same input. The variation demonstrates that at least one additional constitutive relation is required.

B.12 Proposition: Subject-localism is insufficient under worldly resistance

Suppose one subject maintains a constant interpretation while changes in worldly conditions produce different successful consequences.

Then no function:

$$\mu = g(a)$$

is sufficient to account for semantic adequacy.

A subject-only model predicts no relevant semantic difference when the subject state is fixed. Differential worldly consequences demonstrate that external conditions participate in the binding.

B.13 Proposition: Semantic identity is constitution-relative

Let:

$$r_1$$

and:

$$r_2$$

be realizations of concept C.

Then:

$$r_1 \sim_C r_2$$

only relative to a specified invariant set:

$$I_C$$

Without declared invariants, no principled criterion distinguishes identity-preserving variation from semantic replacement.

B.14 Proposition: Global semantic integration is not guaranteed

Let:

$$b_i$$

be locally admissible bindings.

Local admissibility does not entail the existence of a global binding whose restrictions equal every local binding.

Two local bindings may disagree on an overlap, assign incompatible roles, or possess conflicting consequence profiles.

A global section exists only when compatibility and gluing conditions are satisfied.

B.15 Proposition: Semantic location may be non-unique

A meaning type may possess several minimal support graphs.

The same constitutional binding may be realized through speech, writing, gesture, memory, or replicated computational media.

If none of these supports contains another and each is sufficient, the family of minimal loci contains more than one member.

B.16 Proposition: Root callability does not require lexical primitiveness

A concept's root callable kernel may be structurally prior to every lexical term used to invoke it.

Several terms may activate distinct projections of one binding constitution.

If no term alone generates all admissible projections while the shared kernel does, then the callable root is constitutional rather than lexical.

B.17 Proposition: Reflexive Meta-Binding is not self-reference

A system may emit self-referential language without Reflexive Meta-Binding.

If the self-reference does not enter subsequent processing as a causally relevant constraint, the necessary feedback condition is absent.

B.18 Proposition: Binding overextension is unauthorized meta-binding

Let:

$$\widehat{b}$$

be a higher-order binding presented as admitted.

If:

$$\widehat{b} \notin CM_C$$

or its claims exceed its constitutional ceiling, then:

$$\mathrm{BOE}\left(\widehat{b}\right) = 1$$

Binding overextension is therefore not merely factual error. It is unauthorized semantic constitution.

Appendix C: Minimal Semantic Compiler Procedure

C.1 Inputs

The compiler receives:

$$\langle h, c, u, e \rangle$$

where:

- h is a semantic handle;
- c is context;
- u is intended use;
- e is the realization environment.

C.2 Resolution

Generate candidate concept constitutions:

$$\mathcal{C} = \mathrm{ResolveCandidates}_h$$

Select the best supported concept:

$$C^* = \mathrm{SelectConcept}\left(\mathcal{C}, c, u\right)$$

C.3 Kernel retrieval

Retrieve the root callable kernel:

$$K = \mathrm{RetrieveKernel}\left(C^*\right)$$

C.4 Jurisdiction selection

Select active jurisdictions:

$$J^* = \text{SelectJurisdictions}\,(C^*, c, u)$$

Construct the activation vector:

$$\mathbf{z} = \text{ActivationVector}\,(J^*)$$

C.5 Polynomial evaluation

Evaluate:

$$P = \Sigma_{C^*}\,(\mathbf{z})$$

If an interaction term lacks an authorized bridge, return:

$$\perp_{\text{UNSUPPORTED_COMPOSITION}}$$

C.6 Compatibility court

Check:

- role compatibility;
- jurisdiction;
- provenance;
- invariants;
- authority;
- claim ceilings;
- effect feasibility.

If any mandatory check fails, return the corresponding typed refusal.

C.7 Instance Constitution

Create:

$$IC = \text{Constitute}\,(K, P, c, u)$$

C.8 Realization

Produce:

$$IR = \text{Realize}\,(IC, e)$$

C.9 Receipt

Emit:

$$\langle C^*, K, J^*, P, IC, IR, \pi, I_{\text{result}}, C_{\text{result}}, A_{\text{result}} \rangle$$

where the final fields record:

- provenance result;
- invariant result;
- claim-ceiling result;
- authority result.

Appendix D: Binding Hierarchy

The theory distinguishes the following levels.

D.1 Causal coupling

States influence subsequent states.

Causal coupling is necessary for physical realization but is not automatically semantic.

D.2 Pattern binding

Differences are organized into recurrence, similarity, or detectable structure.

D.3 Proto-semantic binding

Signals become linked to regulation, expectation, or action.

D.4 Semantic binding

Relata become mutually determining within a typed, constrained, consequential, and jurisdictional structure.

D.5 Conceptual binding

Many realizations are stabilized into a reusable concept constitution.

D.6 Recursive Meta-Binding

An existing binding becomes a relatum in a higher-order binding.

D.7 Reflexive Meta-Binding

A system binds a representation of its own active binding constitution into subsequent processing.

D.8 Governed Reflexive Meta-Binding

Reflexive operation is constrained by provenance, claim ceilings, authority, and authorized disclosure.

D.9 Epistemic Autonomy

The system possesses authorized capacity to revise some bindings governing its future operation.

The hierarchy may be summarized as:

$$\begin{aligned}\text{causal coupling} &\rightarrow \text{pattern binding}\\ &\rightarrow \text{proto-semantic binding}\\ &\rightarrow \text{semantic binding}\\ &\rightarrow \text{conceptual binding}\\ &\rightarrow \text{recursive meta-binding}\\ &\rightarrow \text{reflexive meta-binding}\\ &\rightarrow \text{governed reflexive meta-binding}\\ &\rightarrow \text{epistemic autonomy}\end{aligned}$$

This is a structural ordering, not a universal chronology.

Higher levels may reach downward and reorganize lower ones.

Corpus-Wide Reference Positioning

This paper argues that where does meaning exist?. Its nearest external comparison fields are semantic, governance, provenance. The sources in [E1], [E2], [E3], [E4], [E5], [E6], [E7] are cited as predecessor literature, adjacent standards, or counterpressure - not as evidence that they caused the Ableman framework or that they validate its novel claims. The comparison isolates where established work already supplies provenance, formal assurance, institutional constraint, empirical method, or engineering practice, and where this paper adds a distinct claim about admissibility, authority, semantic custody, or governed realization.

Within the Ableman corpus, the highest-priority current cross-links are [A1], [A2], [A3], [A4], [A5], [A6], [A7]. These are links among separately custodied works, not silent identity mergers. Their role is to expose mutual dependencies, terminology collisions, and places where a reader should traverse the corpus before treating this paper as self-sufficient.

Reference-pass counterevidence and limit search

The pass applied the following pressure tests: Formal vocabularies and graph links make commitments explicit but cannot by themselves establish correct

identity, shared interpretation, or lossless translation between contexts. A written constitution can legitimate domination as easily as constrain it unless amendment, participation, enforcement, and appeal remain effective in practice. A complete provenance graph can faithfully record a false or unauthorized process; provenance is evidence about lineage, not automatic truth or legitimacy. These limits are not generic disclaimers: they identify the assumptions that a later empirical, formal, or implementation review must try to break.

Reference-pass influence and novelty boundary

No newly added external source is asserted to be a documented authorial influence unless the predecessor source already said so. `Predecessor` means earlier relevant work;

`adjacent` means a useful comparison; `standard` means an interoperability or assurance baseline; and `counterevidence` marks work that narrows or pressures a claim. Novelty therefore remains a reviewable, claim-level question rather than an inference from a low similarity score or an absent citation.

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[**E2**] Alistair Miles and Sean Bechhofer, eds. *SKOS Simple Knowledge Organization System Reference*. W3C Recommendation, 18 August 2009. https://www.w3.org/TR/skos-reference **Relationship:** Adjacent standard for concepts, labels, schemes, notes, and semantic mappings.

[**E3**] W3C OWL Working Group. *OWL 2 Web Ontology Language Document Overview*, Second Edition. W3C Recommendation, 11 December 2012. https://www.w3.org/TR/owl2-overview/ **Relationship:** Adjacent formal ontology standard used to compare declared semantics and inference commitments.

[**E4**] Mark D. Wilkinson et al. "The FAIR Guiding Principles for Scientific Data Management and Stewardship." *Scientific Data* 3, 2016, 160018. DOI: 10.1038/sdata.2016.18. https://doi.org/10.1038/sdata.2016.18 **Relationship:** Predecessor stewardship principles for findability, accessibility, interoperability, and reuse.

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